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APPARATUS FOR AEROSOL GENERATING DEVICE HAVING HEATING CONTROL CIRCUIT HEATING BASED ON TEMPERATURE OF THE HEATER OR HEATERS

Abstract

Apparatus for an aerosol generating device is described. The apparatus comprises a heating circuit for heating a heating arrangement, the heating arrangement arranged to heat an aerosol generating material to thereby generate an aerosol; a temperature sensing arrangement for sensing a temperature of the device; and a controller for controlling a supply of energy to the heating circuit. The controller is configured to: determine a characteristic that is indicative that energy is being supplied to the heating circuit over a given time period, and determine a change in temperature sensed by the temperature sensing arrangement over the given time period; and take a control action if, based on the determined characteristic and the increase in temperature sensed by the temperature sensing arrangement over the given period, the controller determines that one or more predetermined criteria that are indicative of a fault with the temperature sensing arrangement are satisfied.

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Background/Summary

PRIORITY CLAIM [0001] The present application is a continuation of U.S. application Ser. No. 17/593,159 filed Sep. 10, 2021, which is a national phase entry of International Application No. PCT/EP2020/056223, filed Mar. 9, 2020, which claims priority from U.S. provisional Application No. 62/816,287, filed Mar. 11, 2019, each of which is hereby fully incorporated by reference herein.

TECHNICAL FIELD

[0002] The present invention relates to apparatus for an aerosol generating device.

BACKGROUND

[0003] Smoking articles such as cigarettes, cigars, and the like, burn tobacco during use to create tobacco smoke. Attempts have been made to provide alternatives to these articles that burn tobacco by creating products that release compounds without burning. Examples of such products are heating devices which release compounds by heating, but not burning, the material. The material may be for example tobacco or other non-tobacco products, which may or may not contain nicotine.

SUMMARY

[0004] According to a first aspect of the present disclosure, there is provided apparatus for an aerosol generating device, the apparatus comprising: a heating circuit for heating a heating arrangement, the heating arrangement arranged to heat an aerosol generating material to thereby generate an aerosol; a temperature sensing arrangement for sensing a temperature of the device; and a controller for controlling a supply of energy to the heating circuit, wherein the controller is configured to: determine a characteristic that is indicative that energy is being supplied to the heating circuit over a given time period, and determine a change in temperature sensed by the temperature sensing arrangement over the given time period; and take a control action if, based on the determined characteristic and the increase in temperature sensed by the temperature sensing arrangement over the given period, the controller determines that one or more predetermined criteria that are indicative of a fault with the temperature sensing arrangement are satisfied.

[0005] The temperature sensing arrangement may comprise a temperature sensor for attaching to the heating element to sense a temperature of the heating element, and the one or more predetermined criteria are indicative that the temperature sensor is detached from the heating

element.

[0006] The controller may be configured to determine a ratio of the amount of energy supplied to the heating circuit over the given period to the increase in temperature sensed by the temperature sensing arrangement over the given period; and take the control action if the supply of energy to the heating circuit if the ratio equals or exceeds a predetermined value.

[0007] The predetermined value for the ratio may be 2000 mJ/° C. to 6000 mJ/° C., or around 4000 mJ/° C.

[0008] The control action taken by the controller if the one or more predetermined criteria are satisfied may be to adjust the supply of energy to the heating circuit. For example, to stop the supply of energy to the heating circuit or to reduce the supply of energy to the heating circuit.

[0009] The controller may be configured to determine the predetermined criteria for the given time period and to determine the predetermined criteria once for each of one or more further predetermined periods in a usage session of the device, wherein, the predetermined periods may each have a duration of 1/80s to 1/20s or a duration of around 1/64s.

[0010] The heating circuit may be an induction heating circuit, and the heating element may be a susceptor arrangement for being inductively heated by the induction heating circuit, and the temperature sensing arrangement may comprise a temperature sensor for sensing a temperature of the susceptor arrangement.

[0011] The temperature sensor may be a thermocouple for attaching to the susceptor arrangement.

[0012] The temperature sensing arrangement may comprise: a first temperature sensor for measuring a first temperature in the device; and a second temperature sensor for measuring a second temperature in the device; and the increase in temperature sensed by the temperature sensing arrangement over the given period may be an increase in the first temperature or an increase in the second temperature.

[0013] The first temperature may be a temperature of a first heating zone in the device and the second temperature may be a temperature of a second heating zone in the device.

[0014] The heating circuit may be configured to selectively heat the first heating zone and the second heating zone, and the controller may be configured to, during the given period, activate the heating circuit to heat only one of the first heating zone and the second heating zone.

[0015] The controller may be configured to determine the predetermined criteria for the given time period and to determine the predetermined criteria once for each of one or more further predetermined periods in a usage session of the device and during each period the heating circuit may be configured to selectively heat only one of the first heating zone and the second heating zone.

[0016] The increase in temperature sensed by the temperature sensing arrangement and used to determine the one or more criteria for each period during the usage session may be an increase in the first temperature if the heating circuit is active to heat the first heating zone during the period; and an increase in the second temperature if the heating circuit is active to heat the second heating zone during the period.

[0017] The heating circuit may be an induction heating circuit comprising a first inductor coil and a second inductor coil; and the heating element may be a susceptor arrangement, and the first heating zone may be a first zone of the susceptor arrangement arranged in use to be heated by the first inductor coil and the second heating zone may be a second zone of the susceptor arrangement arranged in use to be heated by the second inductor coil.

[0018] The first temperature sensor may be a first thermocouple for attaching to the first zone of the susceptor arrangement and the second temperature sensor may be a second thermocouple for attaching to the second zone of the susceptor arrangement.

[0019] The first thermocouple and second thermocouple may be J-type thermocouples each comprising a constantan wire and an iron wire.

[0020] The first thermocouple may comprise a first constantan wire and the second thermocouple

may comprise a second constantan wire and the first thermocouple and second thermocouple may share a single iron wire.

[0021] According to a second aspect of the present disclosure there is provided an aerosol generating device comprising apparatus according to the first aspect of the present disclosure, wherein the aerosol provision device is for generating an aerosol for being inhaled by a user.

[0022] The device may be a tobacco heating device, also known as a heat-not-burn device.

[0023] According to a third aspect of the present disclosure there is provided an aerosol generating system comprising an aerosol generating device according to the second aspect and an article comprising an aerosol generating material for being heated by the device in use to thereby generate an aerosol.

[0024] Optionally, the aerosol generating material may comprise a tobacco material.

[0025] According to a fourth aspect of the present disclosure there is provided a method for a controller of apparatus for an aerosol generating device, the apparatus comprising: a heating circuit for heating a heating arrangement, the heating arrangement arranged to heat an aerosol generating material to thereby generate an aerosol; a temperature sensing arrangement for sensing a temperature of the device; and the controller, wherein the controller is for controlling a supply of energy to the heating circuit; wherein the method comprises: determining a characteristic that is indicative that energy is being supplied to the heating circuit during a given time period; determining a change in temperature sensed by the temperature sensing arrangement over the given time period; and taking a control action if, based on the characteristic and the increase in temperature sensed by the temperature sensing arrangement over the given period, the controller determines that one or more predetermined criteria that are indicative of a fault with the temperature sensing arrangement are satisfied.

[0026] According to a fifth aspect of the present disclosure there is provided a set of machine-readable instructions which when executed cause the method according to the fourth aspect to be performed.

[0027] According to a sixth aspect of the present disclosure there is provided a machine-readable medium comprising a set of instructions according to the fifth aspect.

[0028] Further features and advantages of the invention will become apparent from the following description of preferred embodiments of the invention, given by way of example only, which is made with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0029] Subject matter hereof may be more completely understood in consideration of the following detailed description of various embodiments in connection with the accompanying figures, in which:

[0030] FIG. 1 shows a front view of an example of an aerosol generating device;

[0031] FIG. 2 shows a front view of the aerosol generating device of FIG. 1 with an outer cover removed;

[0032] FIG. 3 shows a cross-sectional view of the aerosol generating device of FIG. 1;

[0033] FIG. 4 shows an exploded view of the aerosol generating device of FIG. 2;

[0034] FIG. 5A shows a cross-sectional view of a heating assembly within an aerosol generating device;

[0035] FIG. 5B shows a close-up view of a portion of the heating assembly of FIG. 5A;

[0036] FIG. 6 shows a schematic representation of an example induction heating circuit for the aerosol generating device of FIGS. 1 to 5B;

[0037] FIG. 7A shows a schematic representation of a current through an inductor of the example

induction heating circuit of FIG. 6;

[0038] FIG. 7B shows a schematic representation of a voltage across a current sense resistor of the example induction heating circuit of FIG. 6;

[0039] FIG. 8 shows a schematic representation of a voltage across a switching arrangement of the circuit of FIG. 6;

[0040] FIG. 9 shows another schematic representation of the example induction heating circuit for the device of FIGS. 1 to 5B;

[0041] FIGS. 10 to 13 show various parts of an example control arrangement for the example induction heating circuit represented by previous figures;

[0042] FIG. 14 shows a flow chart representation of an example method of controlling aspects of an example induction heating circuit;

[0043] FIG. 15 shows a flow chart representation of another example method of controlling aspects of an example induction heating circuit;

[0044] FIG. 16 shows a schematic representation of a temperature of a susceptor and a target power to be supplied to heat the susceptor throughout an example usage session of an aerosol generating device;

[0045] FIG. 17 shows a perspective view of the susceptor and an example temperature sensing arrangement for measuring a temperature of the susceptor;

[0046] FIG. 18 shows a perspective view of the susceptor and another example temperature sensing arrangement for measuring the temperature of the susceptor;

[0047] FIG. 19 shows example apparatus for providing control functions in the device, where the control functions provided by the example apparatus of FIG. 19 relate to temperatures in the device; and

[0048] FIG. 20 shows a flowchart representation of a method of controlling power supplied to heat the susceptor.

DETAILED DESCRIPTION OF THE DRAWINGS

[0049] As used herein, the term “aerosol generating material” includes materials that provide volatilized components upon heating, typically in the form of an aerosol. Aerosol generating material includes any tobacco-containing material and may, for example, include one or more of tobacco, tobacco derivatives, expanded tobacco, reconstituted tobacco or tobacco substitutes. Aerosol generating material also may include other, non-tobacco, products, which, depending on the product, may or may not contain nicotine. Aerosol generating material may for example be in the form of a solid, a liquid, a gel, a wax or the like. Aerosol generating material may for example also be a combination or a blend of materials. Aerosol generating material may also be known as “smokable material”.

[0050] Apparatuses are known that heat aerosol generating material to volatilize at least one component of the aerosol generating material, typically to form an aerosol which can be inhaled, without burning or combusting the aerosol generating material. Such an apparatus is sometimes described as an “aerosol generating device,” an “aerosol provision device,” a “heat-not-burn device,” a “tobacco heating product device,” or a “tobacco heating device” or similar. Similarly, there are also so-called e-cigarette devices, which typically vaporize an aerosol generating material in the form of a liquid, which may or may not contain nicotine. The aerosol generating material may be in the form of or be provided as part of a rod, cartridge or cassette or the like which can be inserted into the apparatus. A heater for heating and volatilizing the aerosol generating material may be provided as a “permanent” part of the apparatus.

[0051] An aerosol provision device can receive an article comprising aerosol generating material for heating. An “article” in this context is a component that includes or contains in use the aerosol generating material, which is heated to volatilize the aerosol generating material, and optionally other components in use. A user may insert the article into the aerosol provision device before it is heated to produce an aerosol, which the user subsequently inhales. The article may be, for example,

of a predetermined or specific size that is configured to be placed within a heating chamber of the device which is sized to receive the article.

[0052] FIG. 1 shows an example of an aerosol provision device **100** for generating aerosol from an aerosol generating medium/material. In broad outline, the device **100** may be used to heat a replaceable article **110** comprising the aerosol generating medium, to generate an aerosol or other inhalable medium which is inhaled by a user of the device **100**.

[0053] The device **100** comprises a housing **102** (in the form of an outer cover) which surrounds and houses various components of the device **100**. The device **100** has an opening **104** in one end, through which the article **110** may be inserted for heating by a heating assembly. In use, the article **110** may be fully or partially inserted into the heating assembly where it may be heated by one or more components of the heater assembly.

[0054] The device **100** of this example comprises a first end member **106** which comprises a lid **108** which is moveable relative to the first end member **106** to close the opening **104** when no article **110** is in place. In FIG. 1, the lid **108** is shown in an open configuration, however the cap **108** may move into a closed configuration. For example, a user may cause the lid **108** to slide in the direction of arrow "A".

[0055] The device **100** may also include a user-operable control element **112**, such as a button or switch, which operates the device **100** when pressed. For example, a user may turn on the device **100** by operating the switch **112**.

[0056] The device **100** may also comprise an electrical component, such as a socket/port **114**, which can receive a cable to charge a battery of the device **100**. For example, the socket **114** may be a charging port, such as a USB charging port. In some examples the socket **114** may be used additionally or alternatively to transfer data between the device **100** and another device, such as a computing device.

[0057] FIG. 2 depicts the device **100** of FIG. 1 with the outer cover **102** removed. The device **100** defines a longitudinal axis **134**.

[0058] As shown in FIG. 2, the first end member **106** is arranged at one end of the device **100** and a second end member **116** is arranged at an opposite end of the device **100**. The first and second end members **106**, **116** together at least partially define end surfaces of the device **100**. For example, the bottom surface of the second end member **116** at least partially defines a bottom surface of the device **100**. Edges of the outer cover **102** may also define a portion of the end surfaces. In this example, the lid **108** also defines a portion of a top surface of the device **100**. FIG. 2 also shows a second printed circuit board **138** associated within the control element **112**.

[0059] The end of the device closest to the opening **104** may be known as the proximal end (or mouth end) of the device **100** because, in use, it is closest to the mouth of the user. In use, a user inserts an article **110** into the opening **104**, operates the user control **112** to begin heating the aerosol generating material and draws on the aerosol generated in the device. This causes the aerosol to flow through the device **100** along a flow path towards the proximal end of the device **100**.

[0060] The other end of the device furthest away from the opening **104** may be known as the distal end of the device **100** because, in use, it is the end furthest away from the mouth of the user. As a user draws on the aerosol generated in the device, the aerosol flows away from the distal end of the device **100**.

[0061] The device **100** further comprises a power source **118**. The power source **118** may be, for example, a battery, such as a rechargeable battery or a non-rechargeable battery. Examples of suitable batteries include, for example, a lithium battery (such as a lithium-ion battery), a nickel battery (such as a nickel-cadmium battery), and an alkaline battery. The battery is electrically coupled to the heating assembly to supply electrical power when required and under control of a controller (not shown in FIG. 2) to heat the aerosol generating material. In this example, the battery is connected to a central support **120** which holds the battery **118** in place.

[0062] The device further comprises at least one electronics module **122**. The electronics module **122** may comprise, for example, a printed circuit board (PCB). The PCB **122** may support at least one controller, such as a processor, and memory. The PCB **122** may also comprise one or more electrical tracks to electrically connect together various electronic components of the device **100**. For example, the battery terminals may be electrically connected to the PCB **122** so that power can be distributed throughout the device **100**. The socket **114** may also be electrically coupled to the battery via the electrical tracks.

[0063] In the example device **100**, the heating assembly is an inductive heating assembly and comprises various components to heat the aerosol generating material of the article **110** via an inductive heating process. Induction heating is a process of heating an electrically conducting object (such as a susceptor) by electromagnetic induction. An induction heating assembly may comprise an inductive element, for example, one or more inductor coils, and a device for passing a varying electric current, such as an alternating electric current, through the inductive element. The varying electric current in the inductive element produces a varying magnetic field. The varying magnetic field penetrates a susceptor suitably positioned with respect to the inductive element, and generates eddy currents inside the susceptor. The susceptor has electrical resistance to the eddy currents, and hence the flow of the eddy currents against this resistance causes the susceptor to be heated by Joule heating. In cases where the susceptor comprises ferromagnetic material such as iron, nickel or cobalt, heat may also be generated by magnetic hysteresis losses in the susceptor, i.e. by the varying orientation of magnetic dipoles in the magnetic material as a result of their alignment with the varying magnetic field. In inductive heating, as compared to heating by conduction for example, heat is generated inside the susceptor, allowing for rapid heating. Further, there need not be any physical contact between the inductive heater and the susceptor, allowing for enhanced freedom in construction and application.

[0064] The induction heating assembly of the example device **100** comprises a susceptor arrangement **132** (herein referred to as “a susceptor”), a first inductor coil **124** and a second inductor coil **126**. The first and second inductor coils **124**, **126** are made from an electrically conducting material. In this example, the first and second inductor coils **124**, **126** are made from litz wire/cable which is wound in a helical fashion to provide helical inductor coils **124**, **126**. Litz wire comprises a plurality of individual wires which are individually insulated and are twisted together to form a single wire. Litz wires are designed to reduce the skin effect losses in a conductor. In the example device **100**, the first and second inductor coils **124**, **126** are made from copper litz wire which has a substantially circular cross section. In other examples the litz wire can have other shape cross sections, such as rectangular.

[0065] The first inductor coil **124** is configured to generate a first varying magnetic field for heating a first section of the susceptor **132** and the second inductor coil **126** is configured to generate a second varying magnetic field for heating a second section of the susceptor **132**. Herein, the first section of the susceptor **132** is referred to as the first susceptor zone **132a** and the second section of the susceptor **132** is referred to as the second susceptor zone **132b**. In this example, the first inductor coil **124** is adjacent to the second inductor coil **126** in a direction along the longitudinal axis **134** of the device **100** (that is, the first and second inductor coils **124**, **126** do not overlap). In this example the susceptor arrangement **132** comprises a single susceptor comprising two zones, however in other examples the susceptor arrangement **132** may comprise two or more separate susceptors. In some examples, there may be more than two heating zones. Each zone may be formed by respective parts of a single susceptor of the susceptor arrangement or by separate susceptors of the susceptor arrangement. Ends **130** of the first and second inductor coils **124**, **126** are connected to the PCB **122**.

[0066] It will be appreciated that the first and second inductor coils **124**, **126**, in some examples, may have at least one characteristic different from each other. For example, the first inductor coil **124** may have at least one characteristic different from the second inductor coil **126**. More

specifically, in one example, the first inductor coil **124** may have a different value of inductance than the second inductor coil **126**. In FIG. 2, the first and second inductor coils **124**, **126** are of different lengths such that the first inductor coil **124** is wound over a smaller section of the susceptor **132** than the second inductor coil **126**. Thus, the first inductor coil **124** may comprise a different number of turns than the second inductor coil **126** (assuming that the spacing between individual turns is substantially the same). In yet another example, the first inductor coil **124** may be made from a different material to the second inductor coil **126**. In some examples, the first and second inductor coils **124**, **126** may be substantially identical.

[0067] In this example, the inductor coils **124** **126** are wound in the same direction as one another. That is, both the first inductor coil **124**, and the second inductor coil **126** are left-hand helices. In another example, both inductor coils **124**, **126** may be right-hand helices. In yet another example (not shown), the first inductor coil **124** and the second inductor coil **126** are wound in opposite directions. This can be useful when the inductor coils are active at different times. For example, initially, the first inductor coil **124** may be operating to heat a first section of the article **110**, and at a later time, the second inductor coil **126** may be operating to heat a second section of the article **110**. Winding the coils in opposite directions helps reduce the current induced in the inactive coil when used in conjunction with a particular type of control circuit. In one example where the coils **124**, **126** are wound in different directions (not shown) the first inductor coil **124** may be a right-hand helix and the second inductor coil **126** may be a left-hand helix. In another such embodiment, the first inductor coil **124** may be a left-hand helix and the second inductor coil **126** may be a right-hand helix.

[0068] The susceptor **132** of this example is hollow and therefore defines a receptacle within which aerosol generating material is received. For example, the article **110** can be inserted into the susceptor **132**. In this example the susceptor **132** is tubular, with a circular cross section.

[0069] The device **100** of FIG. 2 further comprises an insulating member **128** which may be generally tubular and at least partially surround the susceptor **132**. The insulating member **128** may be constructed from any insulating material, such as a plastics material for example. In this particular example, the insulating member is constructed from polyether ether ketone (PEEK). The insulating member **128** may help insulate the various components of the device **100** from the heat generated in the susceptor **132**.

[0070] The insulating member **128** can also fully or partially support the first and second inductor coils **124**, **126**. For example, as shown in FIG. 2, the first and second inductor coils **124**, **126** are positioned around the insulating member **128** and are in contact with a radially outward surface of the insulating member **128**. In some examples the insulating member **128** does not abut the first and second inductor coils **124**, **126**. For example, a small gap may be present between the outer surface of the insulating member **128** and the inner surface of the first and second inductor coils **124**, **126**.

[0071] In a specific example, the susceptor **132**, the insulating member **128**, and the first and second inductor coils **124**, **126** are coaxial around a central longitudinal axis of the susceptor **132**.

[0072] FIG. 3 shows a side view of device **100** in partial cross-section. The outer cover **102** is again not present in this example. The circular cross-sectional shape of the first and second inductor coils **124**, **126** is more clearly visible in FIG. 3.

[0073] The device **100** further comprises a support **136** which engages one end of the susceptor **132** to hold the susceptor **132** in place. The support **136** is connected to the second end member **116**.

[0074] The device **100** further comprises a second lid/cap **140** and a spring **142**, arranged towards the distal end of the device **100**. The spring **142** allows the second lid **140** to be opened, to provide access to the susceptor **132**. A user may, for example, open the second lid **140** to clean the susceptor **132** and/or the support **136**.

[0075] The device **100** further comprises an expansion chamber **144** which extends away from a proximal end of the susceptor **132** towards the opening **104** of the device. Located at least partially

within the expansion chamber **144** is a retention clip **146** to abut and hold the article **110** when received within the device **100**. The expansion chamber **144** is connected to the end member **106**. [0076] FIG. **4** is an exploded view of the device **100** of FIG. **1**, with the outer cover **102** again omitted.

[0077] FIG. **5A** depicts a cross section of a portion of the device **100** of FIG. **1**. FIG. **5B** depicts a close-up of a region of FIG. **5A**. FIGS. **5A** and **5B** show the article **110** received within the susceptor **132**, where the article **110** is dimensioned so that the outer surface of the article **110** abuts the inner surface of the susceptor **132**. This ensures that the heating is most efficient. The article **110** of this example comprises aerosol generating material **110a**. The aerosol generating material **110a** is positioned within the susceptor **132**. The article **110** may also comprise other components such as a filter, wrapping materials and/or a cooling structure.

[0078] FIG. **5B** shows that the outer surface of the susceptor **132** is spaced apart from the inner surface of the inductor coils **124**, **126** by a distance **150**, measured in a direction perpendicular to a longitudinal axis **158** of the susceptor **132**. In one particular example, the distance **150** is about 3 mm to 4 mm, about 3 mm to 3.5 mm, or about 3.25 mm.

[0079] FIG. **5B** further shows that the outer surface of the insulating member **128** is spaced apart from the inner surface of the inductor coils **124**, **126** by a distance **152**, measured in a direction perpendicular to a longitudinal axis **158** of the susceptor **132**. In one particular example, the distance **152** is about 0.05 mm. In another example, the distance **152** is substantially 0 mm, such that the inductor coils **124**, **126** abut and touch the insulating member **128**.

[0080] In one example, the susceptor **132** has a wall thickness **154** of about 0.025 mm to 1 mm, or about 0.05 mm.

[0081] In one example, the susceptor **132** has a length of about 40 mm to 60 mm, about 40 mm to 45 mm, or about 44.5 mm.

[0082] In one example, the insulating member **128** has a wall thickness **156** of about 0.25 mm to 2 mm, 0.25 mm to 1 mm, or about 0.5 mm.

[0083] As has been described above, the heating assembly of the example device **100** is an inductive heating assembly comprising various components to heat the aerosol generating material of article **110** via an induction heating process. In particular, the first inductor coil **124** and the second inductor coil **126** are used to heat respective first **132a** and second **132b** zones of the susceptor **132** in order to heat the aerosol generating material and generate an aerosol. Below, with reference to FIGS. **6** to **12**, the operation of the device **100** in using the first and second inductor coils **124**, **126** to inductively heat the susceptor arrangement **132** will be described in detail.

[0084] The inductive heating assembly of the device **100** comprises an LC circuit. An LC circuit, has an inductance **L** provided by an induction element, and a capacitance **C** provided by a capacitor. In the device **100**, the inductance **L** is provided by the first and second inductor coils **124**, **126** and the capacitance **C** is provided by a plurality of capacitors as will be discussed below. An induction heater circuit comprising an inductance **L** and a capacitance **C** may in some cases be represented as an RLC circuit, comprising a resistance **R** provided by a resistor. In some cases, resistance is provided by the ohmic resistance of parts of the circuit connecting the inductor and the capacitor, and hence the circuit need not necessarily include a resistor as such. Such circuits may exhibit electrical resonance, which occurs at a particular resonant frequency when the imaginary parts of impedances or admittances of circuit elements cancel each other.

[0085] One example of an LC circuit is a series circuit where the inductor and capacitor are connected in series. Another example of an LC circuit is a parallel LC circuit where the inductor and capacitor are connected in parallel. Resonance occurs in an LC circuit because the collapsing magnetic field of the inductor generates an electric current in its windings that charges the capacitor, while the discharging capacitor provides an electric current that builds the magnetic field in the inductor. When a parallel LC circuit is driven at the resonant frequency, the dynamic impedance of the circuit is at maximum (as the reactance of the inductor equals the reactance of the

capacitor), and circuit current is at a minimum. However, for a parallel LC circuit, the parallel inductor and capacitor loop acts as a current multiplier (effectively multiplying the current within the loop and thus the current passing through the inductor). Allowing the RLC or LC circuit to operate at the resonant frequency for at least some of the time while the circuit is in operation to heat the susceptor may therefore provide for effective and/or efficient inductive heating by providing for the greatest value of the magnetic field penetrating the susceptor.

[0086] The LC circuit used by the device **100** to heat the susceptor **132** may make use of one or more transistors acting as a switching arrangement as will be described below. A transistor is a semiconductor device for switching electronic signals. A transistor typically comprises at least three terminals for connection to an electronic circuit. A field effect transistor (FET) is a transistor in which the effect of an applied electric field may be used to vary the effective conductance of the transistor. The field effect transistor may comprise a body, a source terminal S, a drain terminal D, and a gate terminal G. The field effect transistor comprises an active channel comprising a semiconductor through which charge carriers, electrons or holes, may flow between the source S and the drain D. The conductivity of the channel, i.e. the conductivity between the drain D and the source S terminals, is a function of the potential difference between the gate G and source S terminals, for example generated by a potential applied to the gate terminal G. In enhancement mode FETs, the FET may be OFF (i.e. substantially prevent current from passing therethrough) when there is substantially zero gate G to source S voltage, and may be turned ON (i.e. substantially allow current to pass therethrough) when there is a substantially non-zero gate G-source S voltage.

[0087] One type of transistor which may be used in circuitry of the device **100** is an n-channel (or n-type) field effect transistor (n-FET). An n-FET is a field effect transistor whose channel comprises an n-type semiconductor, where electrons are the majority carriers and holes are the minority carriers. For example, n-type semiconductors may comprise an intrinsic semiconductor (such as silicon for example) doped with donor impurities (such as phosphorus for example). In n-channel FETs, the drain terminal D is placed at a higher potential than the source terminal S (i.e. there is a positive drain-source voltage, or in other words a negative source-drain voltage). In order to turn an n-channel FET “on” (i.e. to allow current to pass therethrough), a switching potential is applied to the gate terminal G that is higher than the potential at the source terminal S.

[0088] Another type of transistor which may be used in the device **100** is a p-channel (or p-type) field effect transistor (p-FET). A p-FET is a field effect transistor whose channel comprises a p-type semiconductor, where holes are the majority carriers and electrons are the minority carriers. For example, p-type semiconductors may comprise an intrinsic semiconductor (such as silicon for example) doped with acceptor impurities (such as boron for example). In p-channel FETs, the source terminal S is placed at a higher potential than the drain terminal D (i.e. there is a negative drain-source voltage, or in other words a positive source-drain voltage). In order to turn a p-channel FET “on” (i.e. to allow current to pass therethrough), a switching potential is applied to the gate terminal G that is lower than the potential at the source terminal S (and which may for example be higher than the potential at the drain terminal D).

[0089] In examples, one or more of the FETs used in the device **100** may be a metal-oxide-semiconductor field effect transistor (MOSFET). A MOSFET is a field effect transistor whose gate terminal G is electrically insulated from the semiconductor channel by an insulating layer. In some examples, the gate terminal G may be metal, and the insulating layer may be an oxide (such as silicon dioxide for example), hence “metal-oxide-semiconductor”. However, in other examples, the gate may be made from other materials than metal, such as polysilicon, and/or the insulating layer may be made from other materials than oxide, such as other dielectric materials. Such devices are nonetheless typically referred to as metal-oxide-semiconductor field effect transistors (MOSFETs), and it is to be understood that as used herein the term metal-oxide-semiconductor field effect transistors or MOSFETs is to be interpreted as including such devices.

[0090] A MOSFET may be an n-channel (or n-type) MOSFET where the semiconductor is n-type. The n-channel MOSFET (n-MOSFET) may be operated in the same way as described above for the n-channel FET. As another example, a MOSFET may be a p-channel (or p-type) MOSFET, where the semiconductor is p-type. The p-channel MOSFET (p-MOSFET) may be operated in the same way as described above for the p-channel FET. An n-MOSFET typically has a lower source-drain resistance than that of a p-MOSFET. Hence in an “on” state (i.e. where current is passing therethrough), n-MOSFETs generate less heat as compared to p-MOSFETs, and hence may waste less energy in operation than p-MOSFETs. Further, n-MOSFETs typically have shorter switching times (i.e. a characteristic response time from changing the switching potential provided to the gate terminal G to the MOSFET changing whether or not current passes therethrough) as compared to p-MOSFETs. This can allow for higher switching rates and improved switching control.

[0091] Now with reference to FIG. 6, circuitry for induction heating by the device **100** will be described. FIG. 6 shows a schematic representation of a part of an induction heating circuit **600** of the aerosol generating device **100**. FIG. 6 shows a part of the induction heating circuit **600** which comprises the first inductor coil **124** for heating the first susceptor zone **132a** when a varying current flows through the first inductor coil **124**. The first susceptor zone **132a** is represented in FIG. 6 as having an inductive element and a resistive element to represent how the susceptor **132** couples inductively with the first inductor **124** and is heated through the generation of eddy currents. It will be noted that the device **100** additionally comprises the second inductor coil **126**, which is not shown in FIG. 6. The second inductor coil **126** is also part of the induction heating circuit **600** and is controlled to heat the second susceptor zone **132b** as will be described below. However, for the sake of clarity, the circuit **600** will first be described with reference to those features shown in FIG. 6.

[0092] The circuit **600** comprises a first resonator section **601**, the DC voltage supply **118** for supplying a DC voltage to the first resonator section **601**, as well as a control arrangement for controlling the circuit **600**. The first resonator section **601** comprises the first inductor **124** and a switching arrangement comprising a first FET **608**, and the control arrangement is configured to switch the FET **608** between a first state and a second state in response to voltage conditions detected in the circuit **600**, as will be described in more detail below, to operate the first inductor **124**. The circuit **600**, with the exception of the susceptor **132**, is arranged on the PCB **122** of the device **100**, with the inductor coil **124** being connected to the PCB **122** at a first end **130a** and a second end **130b**.

[0093] The first resonator section **601** comprises a first capacitor **606**, and a second capacitor **610**, both arranged in parallel with the first inductor **124** such that when the first resonator section **601** is allowed to resonate an alternating current flows between the first capacitor **606** and the second capacitor **610** and through the inductor **124**. As mentioned above, the first FET **608**, in this example an n-channel MOSFET, is arranged to operate as a switching arrangement in the first resonator section **601**.

[0094] It should be noted that in other examples, the resonator section **601** may comprise only one capacitor, for example in the position of the first capacitor **606**, or at the position of the second capacitor **610**. In other examples, the resonator section **601** may comprise any other number of capacitors, such as three or more capacitors. For example, either or both of the first capacitor **606** and the second capacitor **610** may be replaced by two or more capacitors arranged in parallel with one another. As will be well understood, the resonator section **601** has a resonant frequency which is dependent on the inductance L and the capacitance C of the resonator section **601**. The number, type and arrangement of capacitors in the resonating section **601** may be selected based on considerations of the power levels to be used in the circuit **600** and the desired frequency of operation of the circuit **600**. For example, it will be understood that individual capacitors and an arrangement of said capacitors can be considered to have an equivalent series resistance (ESR) as well as a limit on the ability of said capacitors to handle current. Such features may be taken into

account when determining an arrangement of capacitors to provide the capacitance in the resonator section **601**. For example, depending on desired power levels and frequency of operation, there may be an advantage to providing a plurality of capacitors in parallel, to provide higher capacitance or lower ESR. In this example, the first and second capacitors **606**, **610** are both ceramic COG capacitors each having a capacitance of around 100 nF. In other examples, other types of capacitors and/or capacitors with other capacitance values, e.g. capacitors with unequal capacitance values, may be used, according to the considerations outlined in this paragraph.

[0095] The first resonator section **601** is supplied a DC voltage by the DC voltage supply **118**, which is, for example, as described above, a voltage supplied by a battery. As shown in FIG. **6**, the DC voltage supply **118** comprises a positive terminal **118a** and a negative terminal **118b**. In one example, the DC voltage supply **118** supplies a DC voltage of around 4.2V to the first resonator section **601**. In other examples, the DC voltage supply **118** may supply a voltage of 2 to 10V, or around 3 to 5V, for example.

[0096] A controller **1001** is configured to control operation of the circuit **600**. The controller **1001** may comprise a micro-controller, e.g. a micro-processing unit (MPU), comprising a plurality of inputs and outputs. In one example, the controller **1001** is an STM32L051C8T6 model MPU. In some examples, the DC voltage supply **118** provided to the circuit **600** is provided by an output from the controller **1001** which itself receives power from a battery or other power source.

[0097] The positive terminal **118a** of the DC voltage source **118** is electrically connected to a first node **600A**. In an example, the DC voltage source **118** is connected to the node **600A** via the controller **1001** which receives power from the DC voltage source **118** and supplies the voltage supplied by the DC voltage source to components of the device, including the circuit **600**. The first node **600A** is electrically connected to a first end **606a** of the first capacitor **606** and to the first end **130a** of the first inductor **124**. The second end **130b** of the first inductor **124** is electrically connected to a second node **600B**, which in FIG. **6** is represented at two electrically equivalent points in the circuit diagram. The second node **600B** is electrically connected to a drain terminal **608D** of the FET **608**. In this example, the second node **600B** is also electrically connected to a first end **610a** of the second capacitor **610**. Continuing around the circuit, the source terminal **608S** of the first FET **608** is electrically connected to a third node **600C**. The third node **600C** is electrically connected to ground **616**, and in this example to a second end **610b** of the second capacitor **610**. The third node **600C** is electrically connected via a current sense resistor **615** to a fourth node **600D**, and the fourth node **600D** is electrically connected to the negative terminal **118b** of the DC voltage source **118**, which, as with the positive terminal, in an example is supplied via the controller **1001**.

[0098] It should be noted that in examples where the second capacitor **610** is not present, the third node **600C** may have only three electrical connections: to the first FET source terminal **608S**, to ground **616** and to the current sense resistor **615**.

[0099] As mentioned above, the first FET **608** acts a switching arrangement in the first resonator section **601**. The first FET **608** is configurable between a first state, i.e. an 'ON' state and a second state, i.e. an 'OFF' state. As will be well understood by those skilled in the art when an n-channel FET is in an OFF state (i.e. when the appropriate control voltage is not applied to its gate) it effectively acts as a diode. In FIG. **6**, the diode functionality that the first FET **608** exhibits when in its OFF state is represented by a first diode **608a**. That is, when the FET **608** is in the OFF state the first diode **608a** acts to largely prevent current flowing from the drain terminal **608D** to the source terminal **608S** but allows current to flow from the source terminal **608S** to the drain terminal **608D** if the diode **608a** is appropriately forward biased. An n-channel FET is in an ON state when an appropriate control voltage is applied to its gate so that a conductive path exists between its drain D and source S. As such, when the first FET **608** is in the ON state, it acts like a closed switch in the first resonator section **601**.

[0100] As mentioned above, the circuit **600** may be considered to comprise a first resonator section

601 and an additional control arrangement. The control arrangement comprises a comparator **618**, a zero-voltage detector **621**, and a flip-flop **622**, and is configured to detect voltage conditions within the first resonator section **601** and to control the first FET **608** in response to the detected voltage conditions. This control of the first FET **608** by the control arrangement will now be described in more detail.

[0101] At the second node **600B** there is electrically connected the zero-voltage detector **621**, which is configured to detect a voltage condition, i.e. a voltage of at or near 0V with respect to a ground voltage, at a point in the circuit **600** to which the zero-voltage detector **621** is connected. The zero-voltage detector **621** is configured to output a signal to control switching of the state of the FET **608**. That is, the zero-voltage detector **621** is configured to output a signal to the flip-flop **622**. The flip-flop **622** is an electrical circuit which is configurable between two stable states. The flip-flop **622** is electrically connected to a first gate driver **623** which is configured to provide a voltage to the first FET gate terminal **608G** dependent on the state of the flip-flop. That is, the first gate driver **623** is configured to provide an appropriate voltage to the first FET gate terminal **608G** to switch the FET **608** to the ON state when the flip-flop is in one state, but is configured not to provide a voltage appropriate for maintaining the FET **608** in the ON state when the flip-flop **622** is in the other state. For example, the first gate driver **623** may be configured to provide an appropriate gate-source voltage to the first FET gate **608G** to switch the FET **608** ON when the flip-flop **622** is in a state '1', and the first gate driver **623** may be configured not to provide the gate-source voltage when the flip-flop **622** is in state '0'. The state of the flip-flop means **622** therefore controls whether the first FET **608** is on or off.

[0102] In this example, the zero-voltage detector **621** and the first gate driver **623** of the control arrangement are configured to receive respective signals **1011**, **1021** from the controller **1001**, by which signals the controller **1001** can initiate and control operation of the circuit **600**, as will be discussed in more detail below.

[0103] At the fourth node **600D**, there is electrically connected a control voltage line **619**. The control voltage line **619** is electrically connected to a fifth node **600E** via a resistor **617a** and the fifth node **600E** is electrically connected to the voltage comparator **618**—hereinafter comparator **618**. The fifth node **600E** is electrically connected to a positive terminal of the comparator **618**. A negative terminal of the comparator **618** is connected to ground **616**. In this example, the comparator **618** is configured to output a signal based on a comparison of the voltage at the fifth node **600E** to ground voltage. The output signal of the comparator **618** is sent to the flip-flop **622**. A control voltage **1031** is supplied, in this example from the controller **1001**, to the control voltage line **619** via a second resistor **617b**.

[0104] As mentioned above, the comparator **618** is electrically connected to provide an output to the flip-flop **622**. The flip-flop **622** is configured such that an output signal from the comparator **618** can change the state of the flip-flop **622**, and thereby cause the first driver **623** to change the state of the first FET **608**.

[0105] The functioning of the example circuit **600** will now be described in more detail in the context of the first resonator section **601** being activated by the controller **1001** such that the first inductor coil **124** is operated to heat the first susceptor zone **132a**.

[0106] To begin, the first FET **608** is configured in the OFF state, and is thus acting as a diode **608a**, preventing current flowing through the inductor **124**. The controller **1001** initiates the operation of the circuit **600** to heat the first susceptor zone **132a** by causing the FET **608** to switch from the OFF state to the ON state. In this example the controller initiates operation of the circuit **600** by providing a START signal **1011** to the zero-voltage detector **621**. The flip-flop **622** is thereby caused to change states and cause the first gate driver **623** to provide a signal to the FET gate terminal **608G** to thereby switch the FET to the ON state.

[0107] Once the FET **608** is switched to the ON state, what may be referred to as a self-oscillating heating cycle of the circuit **600** begins. The FET **608**, now being in the ON state, acts as a closed

switch allowing a DC current to begin flowing from the DC voltage source positive terminal **118a** through the first inductor **124** and returning to the DC voltage source negative terminal **118b** via the current sense resistor **615**. The first inductor **124** opposes this initial increase in current, as is well-known, generating a back electromotive force (EMF) as understood via Faraday's and Lenz's laws. In the ON state, the voltage between the drain terminal **608D** and the source terminal **608S** is substantially zero.

[0108] FIG. 7A shows a schematic graphical representation of the current flowing through the first inductor **124** against time t starting from when the FET **608** is switched on, at time t_0 . From time t_0 , a DC current begins to build up in the inductor **124** from zero at a rate which is dependent on an inductance $L1$ of the inductor **124**, a DC resistance of the circuit **600** and the DC supply voltage. In one example the current sense resistor **615** has a resistance of around $2\text{ m}\Omega$, while the inductor **124** has a DC resistance of, 2 to $15\text{ m}\Omega$, or 4 to $10\text{ m}\Omega$ or in this example around $5.2\text{ m}\Omega$. This build-up of current in the inductor corresponds to the inductor **124** storing magnetic energy, and the amount of magnetic energy which can be stored by the inductor **124** is dependent on its inductance $L1$, as will be well understood.

[0109] FIG. 7B shows a simplified representation of the voltage across the current sense resistor **615** against time t , again from the time t_0 when the FET **608** is turned on. Shortly after the FET **608** is turned on, a voltage develops across the inductor **124**, this being the back EMF generated by the inductor **124** as the inductor opposes the increase in current. At this time, therefore, the voltage across the current sense resistor **615** as shown in FIG. 7B is small, since almost all of the voltage difference provided by the DC supply **118** drops across the inductor **124**. Then, as the current through the inductor **124** increases and the back EMF of the inductor **124** decays, the voltage across the current sense resistor **615** increases. This is seen as the development of a negative voltage across the current sense resistor **615**, as shown in FIG. 7B. That is, the voltage across the current sense resistor **615** becomes increasingly negative with the length of time that the FET **608** is on.

[0110] Since the increasingly negative voltage across the current sense resistor **615** corresponds with the increasing current through the inductor **124**, the magnitude of the voltage across the current sense resistor **615** is indicative of the current flowing through the inductor **124**. While the FET **608** remains on, the current through the inductor **124** and the voltage across the current sense resistor **615** increase substantially linearly towards respective maximum values I_{max} , V_{max} (which are dependent on the DC voltage supplied by DC supply **118** and the DC resistance of the circuit **600**) with a time constant dependent on the inductance $L1$ and on the DC resistance of the circuit **600**. It should be noted that as the current through the inductor **124** is varying after time t_0 some inductive heating of the susceptor **132** may occur while the DC current through the first inductor **124** builds up.

[0111] The circuit **600** is configured such that the amount of energy which is stored in the first inductor **124** in the time during which the FET **608** is switched on, is determined by the control arrangement and can be controlled by the controller **1001**. That is, the controller **1001** controls an amount of DC current (and thus an amount of magnetic energy) allowed to build up in the inductor **124**, as will now be described.

[0112] As described above, the control voltage **1031** is applied to the control voltage line **619**. In this example, the control voltage **1031** is a positive voltage and the voltage input to the positive terminal of the comparator **618** (i.e. the voltage at the fifth node **600E**) at any one time is dependent on the value of control voltage **1031** and the voltage at the fourth node **600D**. When the negative voltage across the current sense resistor **615** reaches a particular value, it cancels, at the fifth node **600E**, the positive control voltage **1031** and gives a voltage of 0V (i.e. ground voltage) at the fifth node **600E**. In this example, the resistor **617a** has a resistance of $2\text{ k}\Omega$. The resistor **617b** represents an effective resistance to the controller **1001** of $70\text{ k}\Omega$. The voltage at the fifth node **600E** reaches 0V when the negative voltage across the current sense resistor **615** has the same magnitude as the

control voltage **1031**.

[0113] The comparator **618** is configured to compare the voltage at its positive terminal to the voltage of ground **616**, connected to its negative terminal, and output a signal as a result. In one example the comparator is a standard component FAN **156**, as may be obtained from On-Semiconductor. Accordingly, when the voltage at fifth node **600E** reaches 0V, the comparator **618** receives a 0V signal at its positive terminal, and the result of the comparison by the comparator **618** is that the voltage at the positive terminal is equal to the voltage at the negative terminal. The comparator **618** consequently outputs a signal to the flip-flop **622** and causes the FET **608** to be switched off. As such, switching off of the FET **608** is dependent on a voltage condition detected in the circuit **600**. Namely, in this example, when the comparator **618** detects by comparison of the voltage across its terminals that a negative voltage across the current sense resistor **615** has reached the same magnitude as the control voltage **1031**, which occurs at time $t_{sub.1}$, the FET **608** is switched off. In FIG. 7A, the DC current flowing through the inductor **124** at time $t_{sub.1}$ when the FET **608** is switched off is labelled $I_{sub.1}$.

[0114] When the FET **608** is turned off, at time $t_{sub.1}$, the FET **608** switches from acting like a closed switch to acting like a diode **608a** in the resonator section **601**, and for the purposes of supply from the DC supply **118** effectively acting like an open switch. At time $t_{sub.1}$ the path of the DC current through the inductor **124** to ground **616** is interrupted by the FET **608**. This triggers the current flowing in the first inductor **124** to drop off (this is not shown in FIG. 7A), and the inductor **124** opposes this change in current by generating an induced voltage. Accordingly, current begins oscillating back and forth between the inductor **124** and the capacitors **606**, **608** at the resonant frequency of the first resonator section **601**.

[0115] Similarly, the voltage across the inductor **124** and thereby between the first FET drain **608D** and source **608S** terminals begins to oscillate at the resonant frequency of the first resonator section **601**. As the current through and voltage across the inductor **124** begin to oscillate, the susceptor **132** is inductively heated. Switching the FET **608** to the OFF state, therefore acts to release the magnetic energy stored in the inductor **124** at time $t_{sub.1}$ to heat the susceptor **132**.

[0116] FIG. 8 shows a trace **800** of the voltage across the first FET **608**, starting from the FET **608** being in the ON state from time $t_{sub.0}$ to $t_{sub.1}$. Over the time illustrated in FIG. 8 the first FET **608** is turned off and on twice.

[0117] The voltage trace **800** comprises a first section **800a** between times $t_{sub.0}$ and $t_{sub.1}$ when the first FET **608** is ON, and a second section **800b** to **800d** when the first FET **608** is switched off. At **800e** the FET **608** is switched on again, and a third section **800f** which is equivalent to the first section **800a** begins while the first FET **608** remains on and the above-described process of building up of DC current through the inductor **124** repeats. FIG. 8 also shows a fourth section **800g** when the first FET **608** is switched off again to allow oscillation of the voltage across the FET **608**, and a fifth section **800h** when the first FET **608** is subsequently switched on again.

[0118] The voltage across the first FET **608** is zero when the first FET **608** is on in sections **800a**, **800f** and **800h**. When the first FET **608** is turned off as indicated by section **800b** to **800d** and also by section **800g**, the first inductor **124** uses the energy stored in its magnetic field (which magnetic field was the result of the DC current built up when the first FET **608** was on) to induce a voltage that opposes a drop in the current flowing through the first inductor **124** as a result of the first FET **608** being off. The voltage induced in the first inductor **124** causes a corresponding variation in voltage across the first FET **608**. During this variation in voltage, the first inductor **124** and the capacitors **606**, **610** begin to resonate with each other with a sinusoidal waveform. The voltage shown by voltage trace **800** initially increases (see for example **800b**) as the induced voltage in the first inductor **124** increases to oppose a drop in current due to the first FET **608** being off, reaches a peak (see for example **800c**) and then, as the energy stored in the magnetic field of the first inductor **124** diminishes, decreases back to zero (see for example **800d**).

[0119] The varying voltage **800b** to **800d** and **800g** produces a corresponding varying current (not

shown) and, since during the off time of the first FET **608**, the capacitors **606**, **610** and the first inductor **124** act as a resonant LC circuit, the total impedance of the combination of the first inductor **124** and capacitors **606**, **610** is at a minimum during this time. It will therefore be understood that the maximum magnitude of the varying current flowing through the first inductor **124** will be relatively large. This relatively large varying current accordingly causes a relatively large varying magnetic field in the first inductor **124** which causes the susceptor **132** to generate heat. The time period over which the voltage across the first FET **608** varies as indicated by section **800b** to **800d** and by section **800g** in this example depends on the resonant frequency of the first resonator section **601**.

[0120] Referring now to FIG. **6** and FIG. **8**, the circuit **600** is configured such that when the first FET **608** is off and the voltage across the first FET **608** decreases back towards 0V, the zero-voltage detector **621** detects this voltage condition and outputs a signal to the flip-flop **622** which causes the first FET **608** to be switched back to the ON state. That is, in response to this voltage condition detected within the first resonator section **601**, the FET **608** is switched from the OFF state to the ON state. The zero-voltage detector **621** may be considered to detect a voltage condition indicative that a given proportion of a cycle of current oscillation between the inductive element and the capacitive element has been completed since the FET **608** was switched off. That is, the zero-voltage detector **621** detects that a half-cycle of current (and voltage) oscillation at the resonant frequency of the first resonator section **601** has been completed by the zero-voltage detector **621** detecting that the voltage across the FET **608** has returned to 0V or nearly 0V.

[0121] In some examples, the zero-voltage detector **621** may detect when the voltage across the first FET **608** has returned to at or below a voltage level **801** and as such may output a signal to cause switching of the state of the FET **608** before the voltage across the FET **608** reaches exactly 0V. As is illustrated by FIG. **8**, the operation of the zero-voltage detector **621** curtails oscillations of the voltage in the resonator section **601** after one half-cycle and thus results in a substantially half-sine wave voltage profile across the first FET **608**. Further details of the operation of the zero-voltage detector **621** will be described below with reference to FIG. **9**.

[0122] When the first FET **608** is switched back on, at point **800e**, a DC current driven by the DC source **118** again builds up through the first inductor **124**. The first inductor **124** may then again store energy in the form of a magnetic field to be released when the first FET **608** is next switched off to initiate resonance within the first resonator section **601**. As the first FET **608** is repeatedly switched on and off in this way, the above described process is continuously repeated to heat the susceptor **132**.

[0123] It should be noted that the above described building up of current through the inductor **124** described with reference to FIGS. **7A** and **7B** occurs both when the FET **608** is turned on initially in response to a START signal **1011** from the controller **1001** and when the FET **608** is switched on subsequently by a zero-voltage condition detected by the zero-voltage detector **621**. In the first instance, in response to the START signal **1011**, the current in the inductor **124** builds up substantially linearly from 0. In the second instance, when the FET **608** is turned back on in response to a detected zero voltage condition at point **800e**, some excess current is circulating in the circuit **600** (e.g. from previous cycles of switching on and off of the FET **608**). As the FET **608** is turned back on following the detection of a zero-voltage condition, the recirculating current produces an initial negative current through the FET **608**. Then, while the FET **608** remains on, the current through the FET **608** and inductor **124** builds up, substantially linearly, from the initial negative current value produced by the recirculating current. As the current through the inductor **124** builds up, the voltage across the current sense resistor **615** correspondingly becomes increasingly negative, in the manner described above.

[0124] In examples, switching on and off of the FET **608** may occur at a frequency of around 100 kHz to 2 MHz, or around 500 kHz to 1 MHz, or around 300 kHz. The frequency at which the switching on and off of the FET **608** occurs is dependent upon the inductance L, the capacitance C,

the DC supply voltage supplied by the supply **618** and further upon a degree to which current continues recirculating through the resonator section **601** and the loading effect of the susceptor **132**. For example, where the DC supply voltage equals 3.6V, the inductance of the inductor **124** is 140 nH, and the capacitance of the resonator section **601** is 100 nF, the time for which the FET **608** remains on may be around 2700 ns and the time for a half-cycle of oscillation to complete when the FET **608** is off may be around 675 ns. These values correspond to a power of around 20 W being supplied from the DC voltage supply **118** to the resonator section **601**. The above value of the time for which the FET **608** remains on is affected by the amount of current which recirculates in the circuit, since as described above, this recirculating current causes an initial negative current through the inductor upon switching on of the FET **608**. It should also be noted that the time for the current to build up to the value which causes switching off of the FET **608** is also at least in part dependent on the resistance of the inductor **124**, however, this has a relatively minor effect on the time when compared to the effect of the inductance of the resonator section **601**. The time for a half-cycle of oscillation to complete (of in this example 675 ns) is dependent on the resonant frequency of the resonator section **601** which is affected not only by the values of inductance and capacitance of the inductor **124** and capacitors **606**, **610** respectively, but also by the effective resistance provided by loading the inductor **124** with the susceptor **132**.

[0125] Thus far, the circuit **600** has been described in terms of its operation to heat the susceptor **132** by one inductor, the first inductor **124**, and thus only a part of the circuit **600** used by the device **100** has been described. However, as described above, the device **100** also comprises a second inductor **126** for heating the second zone **132b** of the susceptor **132**. FIG. 9 shows the circuit **600** comprising the second inductor **126** in addition to the first inductor **124**.

[0126] As shown in FIG. 9, in addition to the features described with reference to FIGS. 6 to 8, the circuit **600** comprises a second resonator section **701** comprising the second inductor coil **126**, a third capacitor **706**, a fourth capacitor **710** and a second FET **708**, having a drain terminal **708D**, a source terminal **708S**, and a gate terminal **708G**. Additionally, the circuit **600** comprises a second gate driver **723** configured to provide a gate-source voltage to the second FET gate terminal **708G**. The controller **1001** is not shown in FIG. 9 but the controller **1001** controls the circuit **600** in the manner described with reference to FIGS. 6 to 8 and additionally is configured to provide a control signal **1012** to the second gate driver **723**. Some reference numerals of features of the circuit **600** already described with reference to FIG. 6 have been omitted from FIG. 9 for the sake of clarity.

[0127] As described above, the first inductor **124** is arranged to heat the first zone **132a** of the susceptor **132** and the second inductor **126** is arranged to heat the second zone **132b** of the susceptor **132**. The second inductor **126**, third and fourth capacitors **706**, **710**, and second FET **708** are arranged to form the second resonator section **701**, in the same manner as the first inductor **124**, first and second capacitors **606**, **610**, and first FET **608** are arranged to form the first resonator section **601**. In one example, the third and fourth capacitors **706**, **710** are also COG capacitors and may have a capacitance of around 100 nF. The second inductor **126** in one example has a DC resistance of around 8 mΩ. When active, the second resonator section **701** operates to heat the susceptor **132** in an equivalent manner as described above for the first resonator section **601** and description of this will not be repeated here.

[0128] It will be appreciated that the value of the DC resistance of the inductors **124**, **126** will have an effect on the efficiency of the circuit **600**, since a higher DC resistance will result in higher resistive losses in the inductor **124**, **126** and as such it may be desirable to minimize inductor DC resistance, for example by changing the number of windings, or the cross-section of the inductors **124**, **126**. Furthermore, it will be appreciated that an AC resistance of the inductor **124** is increased compared to the DC resistance due to the skin effect. As such, the use of litz wire in examples provides for reducing the skin effect, and thereby reducing AC resistance and associated resistive losses from the inductors **124**, **126**. To give an example, where the first inductor **124** has a DC resistance of around 5 mΩ and the second inductor **126** has a DC resistance of around 8 mΩ, and

the circuit operates at around 300 kHz, the particular arrangement of litz wire forming the coils results in effective resistances for the inductors **124**, **126** of around 1.14 times their DC resistance values.

[0129] A node **700A** in the second resonator section **701** is equivalent to the first node **600A** of the first resonator section **601** and is electrically connected to the first node **600A** and thereby to the positive terminal **118a** of the DC supply **118**. A node **700C** is at the equivalent position in the second resonator section **701** as is the third node **600C** of the first resonator section **601** and the node **700C** is similarly connected to ground **616**.

[0130] It is important to note that the circuit **600** is configured to be operated by the controller **1001** such that only one of the resonator sections **601**, **701** is active at any one time. Examples of this operation will be described in more detail below.

[0131] During the activation of one of the resonator sections **601**, **701**, the zero-voltage detector **621** is configured to detect a zero-voltage condition in the active resonator section **601**, **701** and thus control switching of the respective FET **608**, **708** of the active resonator section **601**, **701**. The zero-voltage detector **621** controls when the respective FET **608**, **708** of the active resonator section **601**, **701**, is switched back on (such as at point **800e**), as will now be described in more detail, with reference to FIGS. **8** to **10**.

[0132] In the circuit **600**, the zero-voltage detector **621** is configured to detect a zero-voltage condition at the second node **600B** of the first resonator section **601** or at the equivalent node **700B** of the second resonator section **701**. When one of the first resonator section **601** and second resonator section **701** is active, the zero-voltage detector **621** detects each time the respective FET **608**, **708** has been switched off, that the voltage across that FET **608**, **708** has returned to zero (e.g. point **800e** in FIG. **8**) or, is close to zero e.g. below a level **801**. In response to the zero-voltage detector **621** making this detection, a signal is output to change the state of the flip-flop **622**. The respective gate driver **623** which is in operation then outputs a gate-source voltage to switch the respective FET back to the ON state.

[0133] A first small signal diode **725** connects the zero-voltage detector **621** to the first resonator section second node **600B** and a second small signal diode **726** connects the zero-voltage detector **621** to the equivalent node **700B** of the second resonator section **701**. Specifically, anodes of the first small signal diode **725** and second small signal diodes are connected to the zero-voltage detector **621** input via a common node **701B** while cathodes of the diodes **725**, **726** are connected respectively to the nodes **600B**, **700B**.

[0134] The operation of the zero-voltage detector **621**, in a particular example, will now be described with reference to FIG. **10**, which shows the zero-voltage detector **621** and the flip-flop **622**. In FIG. **10**, the components which make up the zero-voltage detector **621** are enclosed by a dotted line box. The node **701B** connected to the anodes of the first and second small signal diodes **725**, **726** is shown. The start signal **1011** from the controller **1001** to the zero-voltage detector **621** can also be seen in FIG. **13**.

[0135] The zero-voltage detector **621** in this example comprises an inverter gate **U103** having an input **2** from the node **701B** and an output **4** connected to an input of the flip-flop **622**. The inverter gate **U103** is powered by connections **5** and **3** and a capacitor **C108** isolates the connection **5** from ground. A logic power supply **621a** of, in this example, 2.5V is applied to the input **5** and via a pull-up resistor **R111** to the input **2** of the inverter gate **U103**. The logic power supply **621a** is in this example supplied from the controller **1001**. The inverter gate **U103** is configured to act as an OR gate for the START signal **1011** and a zero-voltage detection signal from the node **701B**. That is, the inverter gate **U103** is configured to receive a logic low signal in the form of the START signal **1011** from the controller **1001** to initiate operation of the circuit **600a**. The START signal **1011** may be provided by on “open drain” signal pin of the controller **1001**. The inverter gate **U103** is also configured to receive a logic low signal from the node **701B** when one of the first and second signal diodes **725**, **726** is forward biased due to one of the nodes **600B**, **700B** being at or near zero

volts, as will be explained below. When either or both of these logic low signals is received by the inverter gate input **2** the inverter gate **U103** inverts the received signal and outputs a logic-high signal to the flip-flop **622**.

[0136] When the first inductor **124** is being operated to heat the susceptor **132**, the second FET **708** remains off. When the second FET **708** remains off, the second small signal diode **726** has either no bias or is reverse biased depending on the voltages at the logic power source and the DC supply **118**, that is, the voltage at a cathode end (nearest the node **700B**) of the second small signal diode **726** is either substantially the same as or higher than the voltage at an anode end (nearest the zero-voltage detector **621**) of the second small signal diode **726**.

[0137] During operation of the first resonator section **601**, when the first FET **608** is off and the voltage across it varies as indicated by **800b-d** of FIG. **8**, the first small signal diode **725** is reverse biased. At the end of this variation in voltage, when the voltage reaches 0V as indicated by **800e**, or is close to 0V (e.g. at or below level **801**), the first small signal diode **725** becomes forward biased. Accordingly, when the first small signal diode **725** is forward biased at **800e**, the signal provided to the input **2** of the inverter gate **U103** becomes a logic low signal since a voltage drop is produced from the logic signal **621a** across the resistor **R111**. As such, once this logic low signal is inverted by the inverter gate **U103**, a logic high signal is provided at the output **4** of the inverter gate **U103**.

[0138] Although in the above description the functioning of the zero-voltage detector **621** is described in relation to controlling switching of the first FET **608**, it will be understood that the zero-voltage detector **621** functions in the same way, using the second small signal diode **726** instead of the first small signal diode **725**, to control the second FET **708**.

[0139] Still with reference to FIG. **10**, the flip-flop **622** comprises a clock input CLK, a reset input/RST, and an output Q. The flip-flop **622** also comprises further inputs D and VCC for supplying power, in this example the flip-flop receives the same 2.5V logic power supply **621a** from the controller **1001** as the inverter gate **U103** receives. The clock input CLK is connected to the output **4** of the inverter gate **U103** and is configured to receive a signal therefrom. When the output **4** of the inverter gate **U103** switches from logic-low to logic-high (due to the input **2** of inverter gate **U103** receiving a detected zero-voltage condition or a START signal **1011** as described above) the clock input CLK of the flip-flop **622** receives a logic-high rising edge signal which “clocks” the flip-flop **622** and makes the state of the flip-flop output Q high. The flip-flop **622** comprises a further input/RST configured to receive a signal from the output of the comparator **618**, by which the comparator **618** can change the state of the flip-flop **621** to cause the flip-flop output Q to be low. The flip-flop output Q is connected to the first and second gate drivers **623**, **723** and on receiving a high output from the flip-flop output Q, whichever one of the gate drivers **623**, **723** is active (due to receiving a signal **1021**, **1022** as described above) provides a gate driver signal to its respective FET **608**, **708**.

[0140] In one particular example, the flip-flop **622** may switch at half of the voltage of the logic power source **621a**, that is, at 1.25V in this example. This means that the forward bias voltage of the first small signal diode **725** and the voltage at the first FET drain **608D** must sum to 1.25V in order that the first FET **608** is switched on. In this example therefore, the first FET **608** is switched on when its drain **608D** is at 0.55V rather than at exactly 0V. It should be noted that ideally, switching may occur at 0V across the FET **608** for maximum efficiency. This zero-voltage switching advantageously prevents the first FET **608** from discharging the capacitors **606**, **610** and thereby wasting energy stored in said capacitors **606**, **610**.

[0141] FIG. **11** shows in more detail the first and second gate drivers **623**, **723** and their connection to the gates **608G**, **708G** of their respective FETs **608**, **708**. Each of the gate drivers **623**, **723** has an input IN which is configured to receive a signal dependent on the heater activation signals **1021**, **1022** supplied from the controller **1001**. Additionally, the signals received by the inputs IN of the gate drivers **623**, **723** are dependent on whether the signal output by the flip-flop output Q is high. The inputs IN are connected to the output Q of the flip-flop **622** via respective resistors **R125**,

R128, which in this example each have a value of 2 k Ω .

[0142] The gate drivers **623**, **723** each have two further inputs VDD and XREF wherein each input VDD receives a 6V power supply from the controller **1001** and XREF receives a 2.5V logic voltage, which in this example is the same logic voltage supplied by the controller **1001** to the flip-flop **622** and inverter gate **U103**. The inputs VDD of each of the first and second gate drivers **623**, **723** are connected to a 6V supply voltage and the inputs VDD are connected to ground via two buffering capacitors **C120**, **C121**. The gate drivers **623**, **723** also each have a terminal GND connected to ground wherein the terminals VDD and GND act to supply power to the gate drivers **623**, **723**. In this example, the capacitors **C120**, **C121** each have a value of 1 μ F. The gate drivers **623**, **723** are configured to output gate drive voltages from respective outputs OUT. The outputs OUT of the gate drivers **623**, **723** are connected respectively to FET gates **608G**, **708G** via resistors **R114**, **R115**, which in this example each have a resistance of 4.992.

[0143] Each gate driver **623**, **723** is configured to receive a signal at its input IN to cause the gate driver to be activated only while a logic-high signal is provided from the flip-flop output Q and a heater activation signal **1021**, **1022** is received from the controller **1001**. An “open-drain” signal pin may be provided on the controller **1001** which is configured to provide the signals **1021**, **1022**.

[0144] In examples, initiation of the circuit **600** for heating by one of the resonator sections **601**, **701** proceeds by the controller **1001** first initiating the desired one of the gate drivers **623**, **723** by a respective one of the heater initiation signals **1021**, **1022**. The controller **1001** then supplies the START signal **1011** to the zero-voltage detector **621**. The duration of the START signal **1011** should be shorter than the period of half a cycle of oscillation by the active resonator section **601**, **701** (this period may be referred to as the “resonant fly-back period”). This allows the circuit to properly begin self-oscillating in response to a detected zero-voltage condition. In another example, the order the START signal **1011** and respective heater enable signal **1021**, **1022** may be reversed such that the START signal **1011** is first applied to set the flip-flop Q output to high, and one of the heater initiation signal **1021**, **1022** then applied to begin the self-oscillating of the resonator section **601**, **701** corresponding to heater to which the signal **1021**, **1022** is supplied.

[0145] To continue with describing in more detail the control arrangement for controlling the circuit **600**, FIG. 12 shows a portion of the control arrangement comprising the comparator **618** and associated components. In FIG. 12, the positive terminal **118a** of the DC power supply **118** is shown connecting to a node **1500A** which is connected to nodes **600A**, **700A** of the first and second resonator sections **601**, **701** respectively. The negative terminal **118b** of the DC power supply is connected to a node **1500B** which is equivalent to the node **600D** shown in FIG. 6. The node **1500B** connects to ground **616** via the current sense resistor **615**. Between the nodes **1500A** and **1500B** an arrangement of capacitors **C111**, **C112**, **C115** and **C116**, each in this example having a capacitance of 100 μ F, are connected in parallel, providing buffering between nodes **1500A** and **1500B**.

[0146] FIG. 12 shows in more detail components associated with the functioning of the comparator **618** for detecting that the current through the active inductor **124** or **126** has reached a given level. As described with reference to earlier figures, the comparator **618** acts to compare a voltage indicative of an amount of DC current flowing in the active inductor (**124** or **126**) to a control voltage **1031** originating from the controller **1001**. The comparator **618** receives power via an input **6** which is connected via a 100 Ω resistor **R116** to a 2.5V logic power signal, in this example supplied by the controller **1001** and the same logic signal as the signal **621a** received by the flip-flop **622**. The comparator power input **6** is connected to ground via a 10 nF capacitor **C119**. A further terminal **2** of the comparator **618** is connected directly to ground.

[0147] In some examples, the controller **1001** is a micro-processing unit comprising a timer (not shown) for generating a signal which produces the control voltage **1031**. In this example, the control voltage **1031** is produced by a pulse-width modulated signal PWM_DAC generated by the controller **1001**. The timer of the controller **1001** generates a pulse-width modulated square

waveform, with, for example, a magnitude of around 2.5V and a frequency of around 20 kHz and having a particular duty cycle. The pulse-width modulated signal PWM_DAC is filtered by 10 nF capacitors C127 and C128, and by two 49.9 k Ω resistors R121, R123 and a 10 k Ω resistor R124 to provide a substantially constant control voltage 1031 at the frequency at which the controller 1001 controls the control voltage 1031 (of, e.g., around 64 Hz in examples). To adjust the control voltage 1031, the controller 1001 in examples is configured to adjust the duty cycle of the pulse-width modulated signal PWM_DAC applied to the circuit 600. As such, the components positioned between the input PWM_DAC and the positive terminal of the comparator 618 effectively provide for the control voltage 1031 to be produced by a pulse-wave modulated signal and for the control voltage 1031 magnitude to be adjusted by adjusting the duty cycle of this pulse wave modulated signal. The control voltage line 619 shown in FIGS. 6 and 9 may thus be replaced by these components. However, in other examples, the control voltage 1031 may be produced by a substantially constant voltage supplied, for example, by the controller 1001. In such examples, some or all of the components for shown in FIG. 12 for filtering the signal PWM_DAC may not be present.

[0148] The node 1500B which is input to the comparator 618 positive input is, as mentioned above, equivalent to the node 600D of the circuit 600. It can be seen from FIG. 12 that, as described with reference to FIG. 6, the node 1500B is connected via the resistor 617a to the positive input of the comparator 618. As such, the operation of the comparator 618 is as described above: to receive an input at its positive terminal which is dependent on the control voltage 1031 and the voltage across the current sense resistor 615. When the voltage at the positive terminal of the comparator 618 reaches ground voltage, a signal /FF RST is output, via a resistor R118, to the flip-flop 622 input/RST to change the state of the flip-flop 622 and thereby switch the active FET 608/708 off.

[0149] FIG. 13 shows further components for a particular example of the control arrangement for the circuit 600. The components shown in FIG. 13 define current sense apparatus 1300 for providing a signal I_SENSE indicative of an amount of current drawn from the DC voltage supply 118 during operation of the circuit 600. From this signal, the controller 1001 may determine a current drawn from the voltage supply 118, and may use this along with a value of the voltage supplied by the DC voltage supply 118 to determine a value for a power supplied to the circuit 600. In some examples, as will be described below, a determined value of power may be used by the controller 1001 for controlling the circuit 600.

[0150] An input 1301 to the current sense apparatus 1300 is provided via the resistor R120 shown in FIG. 12. The input is therefore connected to the node 1500B via the resistor R120 and receives a voltage indicative of the voltage across the current sense resistor 615. The current sense apparatus 1300 operates as a low-side current sensing apparatus for the circuit 600. In that regard, the current sense apparatus 1300 comprises an op-amp U110 running on a voltage of 3.8V supplied to an input 5 of the op-amp U110 (component type TS507) set up for low-side current sensing using the current sense resistor 615, as will be well understood. A transistor U109 with built-in bias resistor (component type RN4986) acts to switch a 2.5V supplied by the controller 1001 up to the 3.8V supply for the op-amp U110. The power supply line from the transistor component U109 is connected to ground via a 10 nF capacitor C132. Further, a 1 k Ω resistor R130 is connected between the positive input of the op-amp U110 and ground and a 412 k Ω resistor R129 is connected between the 2.5V input from the controller 1001 and the positive input of the comparator U110. The negative terminal of the op-amp U110 receives a voltage dependent on the voltage across the current sense resistor 615. A resistor R131 and capacitor C133 in series provide filtering of the voltage signal received via the input 1301. A further resistor R133 (in this example having resistance of 97.6 k Ω) and a 10 nF capacitor C134 are connected in parallel between the input to the negative terminal of the op-amp U110 and the output of the op-amp U110 such that op-amp operates in a closed-loop mode.

[0151] The position of the current sense resistor 615, which, as mentioned above, in one example is

a 2 mΩ resistor, in the circuit allows for a plurality of parameters to be measured with one current sense resistor, which may allow for good efficiency. That is, the position of the current sense resistor **615** in the circuit allows measurement of: the FET peak current, which may be used, for example, in control of the induction heating power of the circuit; the average current out of the battery, which may be used in monitoring discharge of the battery and setting the induction power; and the average current into the battery, which may be used, for example, in monitoring charging of the battery.

[0152] The op-amp **U110** operates to output a voltage signal **I_SENSE** to the controller **1001** which is indicative of the current through the current sense resistor **615** and thus allows the controller **1001** to determine the current drawn from the DC voltage supply **118** through the circuit **600**.

[0153] It should be noted that having regard to the first and second FETs **608** and **708**, and the topology of the circuit **600**, the phasing of the first and second inductor coils **124** and **126** with respect to each other may be chosen such that when the first inductor coil **124** is being operated, current sufficient to cause significant heating of the susceptor **132** is prevented from flowing in the second inductor coil **126**, and when the second inductor coil **126** is being operated, current sufficient to cause significant heating of the susceptor **132** is prevented from flowing in the first inductor coil **124**.

[0154] As described above, the first **608** and second **708** FETs effectively act as diodes **608a**, **708a** when switched off and so may conduct a current if they are forward biased (i.e. the FETs are not perfect switches). Accordingly, in examples the circuit **600** may be configured so that when one of the first **124** and **126** inductor coils is active to heat the susceptor **132**, the voltage induced across the other inactive inductor coil does not forward bias the intrinsic diode of the FET associated with that inactive inductor coil but instead reverse biases it.

[0155] The effect of the above described control arrangement being configured to control the switching arrangements **608**, **708** of the circuit **600** in response to detected voltage conditions is that when one of the first resonator section **601** and the second resonator section **701** is active (i.e. its gate driver **623**, **723** is activated by the controller **1001**) that resonator section “self-oscillates”, while the other section remains inactive. That is, switching of the respective FET **608**, **708** in the resonator section **601**, **701** repeats at a high frequency as a first voltage condition (detected by the comparator **618**) causes the FET to be switched from on to off, and a second voltage condition (detected by the zero-voltage detector **621**) causes the FET to be switched from off to on.

[0156] The controller **1001** is configured to control the induction heating circuit **600** of the device **100** such that only one of the first inductor **124** and the second inductor **126** is active at any one time. The controller **1001** is configured to determine at a predetermined frequency which of the first inductor **124** and the second inductor **126** to activate.

[0157] In examples, during usage of the device **100** the controller **1001** determines at the predetermined frequency, i.e. one time for each of a plurality of predetermined time intervals, which of the first resonator section **601** and the second resonator section **701** to activate. In one example, each time the controller **1001** determines which of the first resonator section **601** and the second resonator section **701** to activate, the controller **1001** may determine to activate that resonator section to heat the susceptor **132** for the duration of the next predetermined interval. That is, where the predetermined frequency (which may be referred to as an “interrupt rate”) is 64 Hz, for example, the controller **1001** may determine at predetermined intervals of 1/64s, which resonator section **601**, **701** to activate for a following duration of 1/64s until the controller makes the next determination of which resonator section **601**, **701**, at the end of the following 1/64s interval. In other examples, the interrupt rate may be, for example, from 20 Hz to 80 Hz, or correspondingly the predetermined intervals may be of length 1/80s to 1/20s. In order to determine which inductor **124**, **126** is to be activated for a predetermined interval, the controller **1001** determines which susceptor zone **132a**, **132b** should be heated for that predetermined interval. In examples, the controller **1001** determines which susceptor zone **132a**, **132b** should be heated with

reference to a measured temperature of the susceptor zones **132a**, **132b**, as will be described below. [0158] FIG. **14** shows a flowchart representation of an example method of determining which of the two resonator sections **601**, **701**, should be activated for a particular predetermined interval. In this example, the controller **1001**, determines which of the first **601** and second **701** resonator sections to activate for the predetermined interval based on a present temperature **T1** of the first susceptor zone **132a** heated by the first inductor **124** and a present temperature **T2** of a second susceptor zone **132b** heated by the second inductor **126**. In an example, the present temperatures **T1** and **T2** of the first **132a** and second **132b** susceptor zones may be measured by respective thermocouples (examples of which are described below) attached to each zone of the susceptor **132**. The thermocouples provide an input to the controller **1001** allowing the controller **1001** to determine the temperatures **T1**, **T2**. In other examples, other suitable means may be used to determine the respective temperatures of the susceptor zones **132a**, **132b**.

[0159] At block **1051**, the controller **1001** determines a present value of the temperature **T1** and compares this to a target temperature **target1** for the first zone **132a** arranged to be heated by the first inductor **124**. The target temperature **target1** of the first zone **132a** has a value which may vary throughout a usage session of the device employing the circuit **600**. For example, a temperature profile may be defined for the first zone defining values for **target1** throughout a usage session of the device **100**.

[0160] At block **1052** the controller **1001** performs the same operation as was performed for the first inductor **124** at block **1051** and determines whether the present temperature **T2** of the second zone **132b** is below the target temperature **target2** for the second zone **132b** at this time. Again, the target temperature of the second zone **132b** may be defined by a temperature profile defining values of **target2** throughout a usage session. The temperature of the second zone **132b** may, similarly to the first zone **132a**, be measured by any suitable means such as by a thermocouple.

[0161] If the answers at block **1051** and block **1052** are both “no”, i.e. both susceptor zones **132a**, **132b** are presently at or above their respective target temperatures **target1**, **target2**, then the controller **1001** determines neither of the first and second resonator sections **601**, **701** should be activated for the next predetermined interval.

[0162] If the answer at block **1051** is “no” and the answer at block **1052** is “yes”, i.e. the first zone **132a** is at or above its target temperature **target1** but the second zone **132b** is below its target temperature **target2**, then the controller **1001** determines that the second resonator section **701** should be activated to heat the second zone **132b** for the next predetermined interval.

[0163] If the answer at block **1051** is “yes” and the answer at block **1052** is “no”, i.e. the first zone **132a** is below its target temperature **target1** and the second zone **132b** is at or above its target temperature **target2**, then the controller **1001** determines that the first resonator section **601** should be activated to heat the first zone **132a** for the next predetermined interval.

[0164] If the answer at block **1051** is “yes” and the answer at block **1052** is “yes”, i.e. both the first **132a** and second **132b** zones are below their respective target temperatures **target1**, **target2**, then the controller **1001** continues to block **1053**. At block **1053** the controller **1001** effectively acts to alternate activation of the first resonator section **601** and second resonator section **701** for each predetermined interval that both zones **132a**, **132b** remain below their respective target temperatures.

[0165] In some examples, in order to alternately activate the first **601** and second **701** resonator sections, at block **1053** the controller **1001** determines if an even number of predetermined intervals has elapsed since the start of the session. If an even number of predetermined intervals has elapsed since the start of the session then the controller **1001** determines that the first resonator section **601** should be activated for the next interval. If an odd number of predetermined intervals has elapsed since the start of the session then the controller **1001** determines that the second resonator section **701** should be activated for the next interval. In other examples, it should be understood, that the controller **1001** may instead activate the second resonator section **701** when an

even number of intervals has elapsed and the first resonator section **601** when an odd number of intervals has elapsed.

[0166] In certain examples, the circuit **600** is configured such that once one of the resonator sections **601**, **701** is activated by receipt of a signal **1021** or **1022** at one of the gate drivers **623**, **624**, the that resonator section **601/701** continues to operate, i.e. self-oscillate, until deactivated by the controller **1001**, for example by providing a different signal to the gate driver of that resonator section **601/701**. As such, upon determining which of the resonator sections **601**, **701** to activate during a given interval, the controller **1001**, in order to initiate this activation may deactivate one of the resonator sections **601**, **701** which was active during a previous interval.

[0167] To illustrate an example of block **1053** where method **1050** shown in FIG. **14** is performed with intervals of 1/64s, if the controller **1001** determines that both zones **132a**, **132b** are below their respective target temperatures target1, target2 and an even number of 1/64s intervals has elapsed since the start of the usage session of the device **100**, then the controller **1001** activates the first resonator section **601** for the next 1/64s interval while the second resonator section **701** is rendered inactive, which in examples requires the controller **1001** deactivating the second resonator section **701**. If after this next interval of 1/64s both zones **132a**, **132b** remain below their respective target temperatures target1, target2, then for the following 1/64s interval the controller **1001** activates the second resonator section **701** while the first resonator section **601** is rendered inactive, which in examples requires the controller **1001** deactivating the second resonator section **701**. For each interval in which both zones **132a**, **132b** remain below their respective target temperatures this alternating between activating the first **601** and second **701** resonator sections continues.

[0168] Altogether, the method **1050** has the effect that the two inductors **124**, **126** are never activated at the same time. Where it is determined that both inductors **124**, **126** require activation to bring their respective zones **132a**, **132b** to target temperature the controller **1001** alternates the supply of power to the inductors **124**, **126** at the predetermined frequency to bring both zones **132a**, **132b** up to their respective target temperature. Therefore, during at a particular point in a usage session, activation of the inductor coils **124**, **126** to heat their respective susceptor zone **132a**, **132b** may be alternating at a particular frequency, such as 64 Hz. It can be seen that, for example, during a period of a usage session comprising a plurality of intervals where the first zone **132a**, is substantially below its target temperature and the second zone **132b** is at or above its target temperature, the method **1050** has the effect that power may be supplied to the first resonator section **601** for close to 100% of this period. However, for a period of a usage session comprising a plurality of intervals in which both zones **132a**, **132b** are below their target temperatures, each inductor may receive power for roughly 50% of this period.

[0169] In examples, the controller **1001** is also configured at predetermined intervals that coincide with the predetermined intervals at which the method **1050** is performed, to determine a power being supplied to one of the resonator sections **601**, **701** from the DC supply **118**.

[0170] As described above, with reference to FIGS. **9** to **11** in particular, in order to control which of the first resonator section **601** and second resonator section **701** is active at any one time, the controller **1001** as well as transmitting a START signal **1001** to initiate operation of the circuit **600** is configured to selectively transmit a first heater operation signal **1011** to the first gate driver **623** to activate the first resonator section **601** or a second heater operation signal **1012** to the second gate driver **723** to activate the second resonator section **701**.

[0171] For example, when the controller **1001** initiates operation of the circuit **600** and the controller **1001** transmits the first heater operation signal **1011**, the circuit **600** operates as described above to activate the first inductor **124** to heat the first susceptor zone **132a**. When the controller **1001** transmits the second heater operation signal **1012** the circuit **600** operates to activate the second inductor **126** to heat the second susceptor zone **132b**. If the controller **1001** transmits neither of the first heater signal **1011** and the second heater signal **1012** then neither inductor **124**, **126** is activated and the susceptor **132** is not heated.

[0172] The controller **1001** is configured to control the power supplied from the DC voltage supply **118** to the circuit **600** for inductive heating of the susceptor **132** based on a comparison of a measurement of power supplied to the circuit **600** and a target power. The controller **1001** is configured to control the power supplied to the circuit **600** by controlling the switching arrangement of the circuit **600**, i.e. by controlling switching of the FETs **608**, **708**. The controller **1001** may control switching of the FETs **608**, **708** by setting the control voltage **1031** which determines the DC current which is allowed to build up in the inductor **124**, **126** corresponding to that FET **608**, **708** before the FET **608**, **708** is switched off.

[0173] FIG. **15** shows an example method **1100** performed by the controller **1001** to control the power supplied to the circuit **600**. At block **1101** the controller **1001** determines the power **P** supplied from the DC supply **118** to the circuit **600**. For example the controller **1001** may determine an average of the power supplied to the circuit **600** during the previous predetermined interval. In examples, the power **P** being supplied to the circuit **600** during an interval may be determined by measurement of the voltage across and DC current being driven through a given one of the resonator sections **601**, **701**. The controller **1001** may then determine the product of the voltage across and DC current through the given one of the resonator sections **601**, **701** to determine the power **P** supplied to that resonator section **601**, **701**.

[0174] In examples, the determined power **P** is an average power supplied from the DC supply **118** over the predetermined interval which may be determined by determining a product of the average DC voltage across the DC supply **118** and the average DC current drawn from the DC supply **118** over the previous interval.

[0175] In the example device **100**, the DC supply **118** is a battery which is connected to the controller **1001**, and the controller **1001** then outputs the voltage of the DC supply **118** to the circuit **600**. The controller **1001** is configured to determine the DC voltage supplied by the battery **118**. The current drawn from the battery **118** is determined by the operation of the current sense apparatus **1300**. The controller **1001** determines the DC voltage and DC current once for every 1/64s interval. The DC voltage can be considered to be essentially constant over this short time period. However, the current is varying at a rate dependent on the rapid rate of switching on and off the circuit. As described above, this is around 300 kHz in some examples. The current sense apparatus **1300** as described above with reference to FIG. **13**, outputs a signal **I_SENSE** which is filtered to remove this around 300 kHz signal. An average DC current for the 1/64s interval is therefore obtained by taking a measurement of this filtered signal **I_SENSE**, and the measurement of **I_SENSE** is taken just before the end of the 1/64s interval in order to allow the signal from the filter to settle. The controller **1001** thereby obtains a DC voltage and DC current measurement for the 1/64s interval and can calculate a product of these values to obtain the determined power **P**. This determined power **P** may be considered to be an average of the power supplied by the DC supply **118** over the 1/64s interval.

[0176] At block **1102** the supplied power **P** determined at block **1101** is compared to a target power. Where the determined power **P** is an average power over the predetermined interval, the target power is a target average power over the same interval. In one example, the target power is a target for the average power supplied over the predetermined interval and may have a value of between 10 and 25 W, or between 15 and 23 W, or around 20 W. In this example, the target power is a range, for example, 20-21 W or of 15-25 W. The controller **1001** may accordingly at block **1102** compare the supplied power value **P** determined at block **1101** to the target range and determine whether the supplied power is below the range, within the target range, or above the target range. For example, where the target range is 20-21 W, at block **1102** the controller **1001** determines whether $P < 20 \text{ W}$, or $20 \text{ W} \leq P \leq 21 \text{ W}$, or $P > 21 \text{ W}$.

[0177] Based on the comparison of the supplied power **P** to the target range, the controller **1001** determines whether and how to adjust the power for the next predetermined interval with the aim of bringing the actual power supplied to the active inductor **124** or **126** during the next predetermined

interval towards the target power range. That is, if the supplied power P is below the target range, then the controller **1001** determines to increase the power supplied to the circuit **600** over the next predetermined interval. If the supplied power P is above the target range, then the controller **1001** determines to decrease the power supplied to the circuit **600** over the next predetermined interval. If the supplied power P is below the target range, then the controller **1001** determines not to adjust the power supplied to the circuit **600** over the next predetermined interval.

[0178] Due to the configuration of the circuit **600** described above, the supplied power P for a given predetermined interval is dependent on the value of the control voltage **1031** for that interval. Taking the example of one $1/64s$ interval for which the first resonator section **601** is active, this $1/64s$ interval comprises many repeating cycles comprising sections **800a** to **800e** of the voltage trace **800** and repeats thereof. For each cycle during the period of time $t_{sub.1}$ to $t_{sub.0}$ the resonator section **601** is allowed to resonate, and since for this period the FET **608** is off, no power is drawn from the DC supply **118** through the first resonator section **601**. Substantially all of the power drawn from the DC supply **118** during the given $1/64s$ interval to power the resonator section **601** is thus drawn during the period between $t_{sub.0}$ and $t_{sub.1}$ while the inductor **124** is being “energized” with current, i.e. while the FET **608** is on. The time between $t_{sub.1}$ and $t_{sub.0}$ is determined by the resonant frequency of the first resonator section **601**. This resonant frequency may remain substantially constant, at least throughout a given $1/64s$ interval (although may vary over the period of operation of the circuit **600** due to dependence on coil and susceptor temperature and battery voltage). The length of time $t_{sub.0}$ to $t_{sub.1}$ is determined by the value of the control voltage **1031**, as well as the DC voltage supplied by the DC supply **118** and the resistance and inductance of the first resonator section **601** (the same applying for the second resonator section **701**). That is, for a given DC supply voltage, the control voltage **1031** sets the current $I_{sub.1}$ which is allowed to build up in the inductor **124** between $t_{sub.0}$ and $t_{sub.1}$, but where the DC supply voltage is reduced, the time required to build up a given value of $I_{sub.1}$ is increased. As such, the average power supplied during the $1/64s$ interval is dependent on the value of the control voltage **1031**.

[0179] In examples, therefore, in order to control the power supplied to the circuit **600** during the next interval the controller **1001** sets the value of the control voltage **1031** for the next interval. In examples, for a given DC supply **118** over a predetermined interval during which one of the resonator sections **601**, **701** is active, a larger positive value of the control voltage **1031** results in a larger value of power P being delivered to the circuit **600**. Therefore, in such examples, where the controller **1001** determines that the supplied power P over the last interval was above the target range, the controller **1001** reduces the control voltage **1031** for the next interval. Where the controller **1001** determines that the supplied power P over the last interval was below the target range, the controller **1001** increases the control voltage **1031** for the next interval. And, where the controller **1001** determines that the supplied power P over the last interval was above the target range, the controller **1001** leaves the control voltage **1031** unchanged for the next interval.

[0180] It should be noted that in one example of the above method **1100** the power supplied P is determined at block **1101** is a power supplied to a particular one of the resonator sections **601**, **701**. For example, the power P may be determined by measuring the voltage across the first resonator section **601** and the DC current through the first resonator section **601**. In such an example, it is the power P supplied to the first resonator section **601** which is used to control the control voltage **1031**. It should also be noted that for a given control voltage **1031**, in some examples, the power supplied to each of the inductors **124**, **126** when the respective resonator sections **601**, **701** are active may be different. This may be, for example, because the inductors **124**, **126** have different values of inductance or DC resistance, or the capacitance of the two resonator sections **601**, **701** is not equal. Therefore, in this example, during a given predetermined interval, a target power outside of the target power range may be supplied to the second resonator section **701** but since the control voltage **1031** is controlled based on the power P supplied to the first resonator section **601**, in this

example the controller **1001** may not adjust the control voltage **1031**.

[0181] For example, for a given value of the control voltage **1031**, the controller **1001** may determine at block **1101** that an average power of 20 W was supplied to the first resonator section **601** over a given interval, with the target voltage in this example being 20-21 W. At block **1102** the controller **1001** determines that the supplied voltage was within the target range and accordingly the controller **1001** determines not to adjust the control voltage **1031**. Consider that for the next predetermined interval, the controller **1001** determines (by the example method **1050**) that the second resonator section **701** and not the first resonator section **601** is to be activated. For the given value of the control voltage **1031**, in this example, 22.5 W is delivered due to differences in the electrical properties of the first **601** and second **701** resonator sections. However, in this example, at block **1102** the controller **1001** compares the last measured value of power P delivered to the first resonator section **601** and therefore determines at block **1103** not to adjust the control voltage **1031**. As such, in an example of the method **1100** the actual power supplied to the circuit **600** may be outside of the target range. However, this may allow for controlling the power supplied to the inductors **124**, **126** by measuring only the power P supplied to one of the resonator sections **601**, **701**. This may provide a simple and useful solution to maintain the power supplied to the circuit **600** to within an acceptable range if, for example, the resonator sections **601**, **701** and components thereof have roughly similar electrical properties.

[0182] As mentioned above, in some examples, the DC supply **118** is a battery with a voltage of around 2 to 10V, or 3 to 5V, or in one example around 4.2V. In some examples, the DC voltage produced by the DC supply **118** may change, e.g. decrease, over the time that the circuit **600** is operated. For example, where the DC voltage source **118** is a battery, the battery may initially supply a voltage of 4.2V but the voltage supplied by the battery may reduce as the battery depletes. After a given period, therefore, the DC voltage source **118** may supply, for example, 3.5V instead of an initial 4.2V.

[0183] As described above, at a given supply voltage, the value of the control voltage **1031** controls the amount of current which is allowed to build up in the active inductor **124/126** before the respective FET **608/708** is switched off. Power is supplied from the DC voltage supply **118** to “energize” the active inductor **124/126** by allowing a build-up of DC current when the FET **608**, **708** is on. As was also described above, the time $t_{sub.1}$ for the current to build up to the value which causes switching of the FET **608/708** is dependent on the DC voltage supply. Therefore, for example, if the voltage supplied by the DC supply **118** reduces, the rate at which current builds up in the inductor coil **124** reduces, resulting in reduced power P being supplied to the circuit **600**.

[0184] The example method **1100** may provide for a target power to be maintained even in the event that the supplied voltage from the DC supply **118** changes. That is, since an actual supplied power P is determined and used to control the control voltage **1031**, the controller **1001** can act to maintain a target power by adjusting the control voltage **1031**. For example, where the battery level has depleted, the controller **1001** measures that the power P supplied to the circuit **600** at a given control voltage **1031** has reduced, and acts to increase the power P supplied to the circuit by increasing the control voltage **1031**. As such, a target power level may be maintained while a battery used to power the circuit **600** depletes. This is advantageous since maintaining a target power level may provide for optimal efficiency of operation of the induction heating circuit **600**. For example, maintaining a substantially constant power supplied allows for consistent heating of the aerosolizable material **110a** regardless of supply voltage. Similarly, the example method **1100** provides for providing a substantially constant power regardless of other changing factors in the circuit which might affect the amount of power delivered, such as different loading on the circuit **600** being provided by the susceptor **132** when the susceptor **132** temperature increases. This provides a consistently good experience for the consumer, for example by providing a consistent time to first puff, i.e. a consistent time between the device **100** being activated and being ready to provide aerosol to be inhaled by the user.

[0185] In another example, the measured power value P upon which control of the control voltage **1031** is based is changed throughout a usage session. For example, during a particular usage session, for a first part of the usage session (e.g. a first ~60 s of the usage session), the temperature profile may be such that the first inductor **124** is primarily active, while the second inductor **126** is inactive. For this first part of the usage session it may be appropriate to base control of the control voltage **1031** on measurements of the power delivered to the first resonator section **601**. However, later in the session, again e.g. due to the temperature profile for the session, it may be that the second inductor **126** is primarily active, while the first inductor **124** is active for less of the time. Thus, for a second part of the usage session (e.g. after ~60 s), it may be advantageous to control the control voltage **1031** based on the measurements of power delivered to the second resonator section **701**. The controller **1001** may accordingly switch from basing control of the control voltage **1031** on measurements of power supplied to the first resonator section **601** to basing control of the control voltage **1031** on measurements of power supplied to the second resonator section **701**. In this way, the target power may be more closely adhered to throughout a usage session, since, for example, the control voltage **1031** is being set based on a comparison of the actual power being delivered to the active inductor **124**, **126** to the target power range.

[0186] In some examples, where the controller **1001** determines at block **1103** that the power should be adjusted, the controller **1001** may adjust the control voltage **1031** in predetermined steps. For example, the controller **1001** may be configured to adjust the control voltage **1031** by a predetermined amount per predetermined time interval. Where at block **1102** the controller **1001** determines that the supplied power P was below the target power range the controller **1001** may increase the control voltage **1031** by a predetermined number of volts for the next predetermined interval. Conversely, where at block **1102** the controller **1001** determines that the supplied power was above the target power range the controller **1001** may increase the control voltage **1031** by a predetermined amount for the next predetermined interval.

[0187] In the example described above with reference in particular to FIG. 12, the control voltage **1031** is produced by a pulse-wave modulated signal PWM_DAC. The signal PWM_DAC, as described above has a rectangular waveform at 2.5V. The duty signal of the signal PWM_DAC is controllable by the controller **1001** which sets a value of 0 to 800 for the PWM_DAC duty cycle, this value corresponding to a duty cycle of 0% at 0 and 100% at 800. The signal PWM_DAC when filtered provides the substantially constant control voltage **1031** and therefore the settings of from 0 to 800 of the duty cycle of the PWM_DAC signal provide for the control voltage **1031** to have a magnitude of from 0 to 2.5V. In this example, the controller **1001** may adjust the duty cycle setting of the PWM_DAC signal by a set amount, such as 8 out of 800, or leave the setting unchanged, for each predetermined interval. In another example, the controller **1001** may provide for the control voltage **1031** to be adjusted by some other means, and if the controller **1001** determines that the control voltage **1031** should be adjusted, the controller **1001** may adjust the control voltage **1031** by, e.g., 1%, or 2%, or 5% of the maximum value of the control voltage **1031** for the next predetermined interval.

[0188] In some examples, when operation of the circuit **600** is initiated by the controller **1001**, e.g. to start a use session of the device **100** comprising the circuit **600**, the control voltage **1031** is set to a predetermined initial value. In one example, a value of the control voltage **1031** (for example, a duty cycle setting of the signal PWM_DAC which produces this value of the control voltage **1031**) which corresponds with a target power level is determined during setup of the circuit **600**. That is, the power delivered to the circuit **600** may be determined (e.g. measured or determined theoretically) for a number of values of the control voltage **1031**, for example to produce a calibration curve. A value of the control voltage **1031** corresponding to the target power may then be determined. In one example, the DC supply **118** may supply 4.2V and to achieve a target power of 20 W the controller **1001** may determine in an example calibration a value for the duty cycle of the PWM_DAC signal setting of around 344 out of 800.

[0189] In one example, the controller **1001** is configured to set the control voltage **1031** at an initial value which is based on this determined value of the control voltage **1031**. For example, the initial value of the duty cycle of PWM_DAC which determines the control voltage **1031** may be set at half of the determined value corresponding to the target power. For example, where the duty cycle setting for the control voltage **1031** found to correspond with the target power is 344 out of 800, the controller **1001** may begin the session with the setting being set at 152 out of 800, and increase the setting by a predetermined amount with every predetermined interval until the measured power P is within the target range. This may have the effect that at the start of a usage session, the power delivered is well below the target power and the power delivered may then ramp up (by ramping up by the controller **1001** of the control voltage **1031**) until it reaches the target power range. This initial ramping up of the power delivered may provide for improved safety in operation of the circuit **600**, preventing overheating of the susceptor at the start of a session and allowing the circuit **600** to respond to the actual power supplied as determined by the controller **1001**.

[0190] In one example, the predetermined interval is the same predetermined interval as is used by the controller **1001** in the method **1050** of determining which of the first **124** and second **126** inductors to activate. In one such example, as mentioned above, the predetermined intervals are of length 1/64s. The length of the predetermined interval (or equivalently the interrupt rate) may be chosen to provide an advantageous time interval at which the controller can monitor the circuit and adjust parameters accordingly. For example, an interrupt rate of 64 Hz, or within a range of approximately 10-100 Hz may be used. At these example interrupt rates, the controller **1001** may measure increases in temperature of the susceptor zones at a sufficiently high rate that it may determine to stop heating by a particular inductor **124**, **126** before a zone **132a**, **132b** of the susceptor **132** can increase too far above its target temperature. Similarly, examples given for the interrupt rate may provide an advantageous frequency at which the control voltage **1031** may be adjusted to allow appropriate control of power supplied to the inductors **124**, **126** to within a safe target range.

[0191] In an example method of operation of the circuit **600**, a target power for use by the controller **1001** in controlling power delivered to the circuit **600** is predetermined based on characteristics of a planned usage session. For example, the target power range may be adjusted throughout a usage session.

[0192] FIG. **16** shows a schematic example of a temperature profile target1 for a portion of a usage session, which in this example is a target temperature for the single susceptor zone **132a**. In this example, initially at a first part **1201** of the portion of the usage session, the first zone **132a** is substantially below its target temperature target1. At this first part **1201**, the circuit **600** is operating to bring the first zone **132a** up to the target temperature target1. At such an example part of the usage session, a target power P1 may have a range of values of, for example, 20-21 W. The target power during the first part **1201** of the session may be relatively high in order to bring the susceptor **132**, and therefore the aerosolizable material **110a**, up to temperature quickly to a temperature suitable for producing aerosol for inhaling by the user.

[0193] As the usage session progresses, the first zone **132a** substantially reaches its target temperature target1. A second part **1202** of the usage session may be defined beginning shortly after the first zone **132a** reaches its target temperature target1. For instance, for this part **1202** of the usage session, the first zone **132a** may be substantially at its target temperature target1, of e.g. 250° C., and may be being maintained at the target temperature target1 according to the method **1050**. Similarly, although this is not shown in FIG. **16**, the second zone **132b** may be being maintained at its own target temperature target2 by the method **1050** (and the target temperature target2 of the second zone **132b** may define a different temperature profile to that defined by target1).

[0194] The part **1202** in the usage session after the first zone **132a** substantially reaches temperature target1 may be characterized in that the controller **1001** is operating to maintain the temperature of the first zone **132a** (or of both zones **132a**, **132b**) rather than to bring the

temperature of the first zone **132a** up to its target value **target1**, as in the first part **1201**. As such, during the part **1202** of the usage session, relatively little power may be required to be supplied to the susceptor zone **132a** to maintain the target temperature **target1**, when compared to the power required to bring the susceptor zone **132a** up to the target temperature **target1**. At the second part **1202** of the usage session, it may be advantageous to reduce the value of target power **P1** compared to its value in part **1201**. In one example, the target power level **P1** may be reduced from 20-21 W in part **1201** to around 15 W during part **1202** of the usage session. Reducing the target power **P1** in this way may be advantageous in some examples because by using a lower level of power energy losses in the circuit may be reduced, and thus efficiency may be increased.

[0195] For a third part **1203** of the usage session, the value of the target temperature **target1** is 0, i.e. the first inductor **124** is not to be activated. At this point, the target power **P1** may also be reduced to 0 if the usage session has come to an end, or if the second inductor **126** is still being activated, then the target power **P1** may remain at a non-zero value while the second inductor **126** is activated. Accordingly, the target power may take into account the temperature profile of both zones **132a**, **132b** at any one point in the usage session. If a part of the usage session, for example, requires one of the zones to be significantly increased in temperature, then a relatively high target power may be appropriate. Conversely, for parts of a usage session where neither zone **132a**, **132b** requires substantial heating, a relatively low target power may be used.

[0196] As mentioned above, use of lower power levels during certain periods of a usage session may provide advantages in that an energy saving may be achieved over the duration of a session. For example, where the target power level is reduced from 20-21 W in the first period to around 15 W in the second period, in some examples an energy saving of around 5-10% may be achieved due to reduced energy losses in the circuit **600** when operating at lower power. In one example, over the course of a typical session of around 260 s in length, maintaining the target power at around 20 W for the full duration of the session may result in energy usage of around 1000 J. However, reducing the target power to around 15 W upon the first zone **132a** first reaching its set temperature and maintaining the target power level at 15 W for the remainder of a session of substantially the same length may result in an energy usage of between 900 and 950 J. In examples, almost all of the power used by the device is due to energy supplied to heat the susceptor **132**. The power usage of electrical components other than the heating circuitry, e.g. LED indicators and the microcontroller, may be less than around 0.1 W and in some examples may be less than around 0.01 W.

[0197] The example device **100** comprises a temperature sensing arrangement for sensing a temperature of the susceptor **132**. For example, the temperature sensing arrangement may comprise one or more temperature sensors, and in one example comprises one temperature sensor for each zone of the susceptor **132**. In one example, as described above, the susceptor comprises a first zone **132a** and a second zone **132b** and the controller **1001** operates the induction heating circuit **600** of the device **100** as described above with reference to earlier figures to heat each zone by a respective inductor **124**, **126**.

[0198] In examples, the controller **1001** is configured to determine whether one or more criteria that are indicative of a fault with the temperature sensing arrangement are satisfied. If the controller determines that said one or more predetermined criteria are satisfied then the controller **1001** takes a control action, for example stopping or reducing the supply of energy to heat the susceptor **132** or issuing a warning signal indicating that a fault has occurred. Examples of said predetermined criteria are described below. A safety feature may therefore be provided which allows the controller **1001** to take a control action if a fault with the temperature sensing arrangement is identified.

[0199] FIG. 17 shows further details of the susceptor **132**. The susceptor **132**, as described above, comprises a first zone **132a**, and a second zone **132b**. A first thermocouple **133a** is arranged to measure the temperature of the first susceptor zone **132a** and a second thermocouple **133b** is arranged to measure the temperature of the second susceptor zone **132b**. As will be well understood, a thermocouple is a device used for sensing temperature which comprises two

dissimilar electrical conductors. The two conductors are held at the same electric potential at a first measurement end while a second end of the conductors is held at a known temperature to form a second reference end. In some examples, the ends of the two wires may be connected at the measurement end or in other examples the two wires may be connected to a single conductive surface, such as that of the susceptor **132**. A voltage is generated between the conductors which is dependent on a temperature difference between the measurement end and the reference end, according to the Seebeck effect. If the temperature of the reference end is known for example through measurement by a temperature sensor, and in one example herein by use of 10 k Ω thermistor, then a temperature at the measurement end can be determined from the voltage generated between the conductors.

[0200] In the example of FIG. **17**, the first thermocouple **133a** comprises a first pair of wires **1704**, **1705** and the second thermocouple **133b** comprises a second pair of wires **1708**, **1709**. In one example, both thermocouples are J-type thermocouples, that is, a first wire **1704**, **1708** of each pair is formed of constantan while a second wire **1705**, **1709** of each pair is formed of iron. In other examples, different types of thermocouples comprising different pairs of dissimilar conductors may be used, such as type E, K, M thermocouples for example, or a different type of thermocouple may be used for each susceptor zone **132a**, **132b**.

[0201] The first thermocouple **133a** comprises a first measurement junction **1706** at which the constantan wire **1704** and iron wire **1705** of the first thermocouple **133a** are joined together and which is attached to the first susceptor zone **132a**. In this example, the first measurement junction **1706** is attached to the first susceptor zone **132a** by spot-welding to the surface of the susceptor **132**. Similarly, the second thermocouple **133b** comprises a second measurement junction **1310** at which the constantan wire **1708** and iron wire **1709** of the first thermocouple **133a** are joined together and which is spot-welded to the susceptor **132** at the second susceptor zone **132b**. Insulating sheaths **1707**, **1711** cover each wire of the first thermocouple **133a** and second thermocouple **133b**. The first thermocouple **133a** terminates at a pair of terminals **1704a**, **1705b** for providing reference end voltages for the wires **1704**, **1705** respectively to the controller **1001**, allowing determination of the temperature at the first measurement junction **1706** by the controller **1001**. Similarly, the second thermocouple **133b** terminates at a pair of terminals **1708a**, **1709a** for providing reference end voltages to the controller **1001** for the wires **1708**, **1709** respectively and allowing determination of the temperature at the second measurement junction **1710** by the controller **1001**. Each of the terminals **1704a**, **1705a**, **1708a**, **1708a** may be connected to respective input pins of the controller **1001**, or may be attached to the PCB **122** and thereby to the controller **1001**. For example the terminals **1704a**, **1705a**, **1708a**, **1708a** may each be soldered to the PCB **122**, to provide an input to the controller **1001**.

[0202] Another example of a temperature sensing arrangement for the susceptor **132** is shown in FIG. **18**. In this example the temperature sensing arrangement again comprises two thermocouples each comprising a constantan wire. However, in this example the two thermocouples share an iron wire. That is, a first thermocouple **183a** is for allowing the controller **1001** to determine a temperature of the first susceptor zone **132a** and comprises a first constantan wire **1804** attached to the first susceptor zone **132a** at a measurement end **1806**. An iron wire **1805** forms a second wire of the first thermocouple **183a** and is attached at the second susceptor zone **132b** in this example. A second thermocouple **183b** for measuring the temperature of the second susceptor zone **132b** comprises a first constantan wire **1808** attached to the second susceptor zone **132b** at a point **1810**. The iron wire **1805** also forms the second wire of the second thermocouple **132b**. The first constantan wire **1804** of the first thermocouple **132a** comprises a reference end **1804a**, the first constantan wire **1808** of the second thermocouple **132b** comprises a reference end **1808a**, and the iron wire **1805** forming the second wire of both the first and second thermocouples **183a**, **183b** comprises a reference end **1805a**. The reference ends **1804a**, **1805a**, **1808a** provide respective reference end voltages for each of the wires **1804**, **1805**, **1808** to the controller **1001** to allow the

controller **1001** to determine temperatures of the first and second susceptor zones **132a**, **132b**. That is, the reference voltages provided by the constantan wire reference end **1804a** of the first thermocouple **183a** and the reference voltage provided by the iron wire reference end **1805a** allow the controller **1001** to determine the temperature of the first susceptor zone **132a**. Similarly, the reference voltages provided by the constantan wire reference end **1808a** of the second thermocouple **183b** and the reference voltage provided by the iron wire reference end **1805a** allow the controller **1001** to determine the temperature of the second susceptor zone **132b**. As described with reference to the example shown in FIG. **17**, the junction ends **1804a**, **1805a**, **1808a** of the wires may be attached to the PCB **122** to provide reference voltages to the controller **1001**. In the arrangement shown in FIG. **18**, the susceptor **132** is iron. This allows for the use of the common iron reference wire **1805**, since, because the iron wire **1805** and the susceptor **132** are made of the same material, the point at which the iron wire **1805** joins to the susceptor **132** is not a thermocouple junction. Providing three wires, as in the arrangement shown in FIG. **18**, may be advantageous when compared to providing four wires, as in the arrangement of FIG. **17**, in that it requires fewer wires to be attached, e.g. soldered, to the PCB **122**.

[0203] FIG. **19** shows a schematic representation of apparatus **1900** for providing control functions relating to temperatures in the device **100**. As can be seen from FIG. **19**, the apparatus **1900** is arranged to receive the reference voltages provided by the reference ends **1804a**, **1805a**, **1808a** of the thermocouple wires **1804**, **1805**, **1808** shown in the temperature sensing arrangement example of FIG. **18**. Describing first the input from the constantan wire **1804** of the first thermocouple **183a**, the constant wire reference end **1804a** of the first thermocouple **183a** is attached to the apparatus **1300** at a point **P8**. From there, the reference voltage signal at **1804a** is arranged to be supplied to the positive and negative input terminals of a first thermocouple op-amp **U4A** (component type OPA2376). The point **P8** is connected to the positive terminal of the op-amp **U4A** via a 2.49 k Ω resistor **R26** and a 2.2 nF capacitor **C20**. The point **P8** is connected to the negative terminal of the op-amp **U4A** via the resistor **R26** and a second 2.49 k Ω resistor **R27**. The op-amp **U4A** is connected in a closed-loop configuration with 536 k Ω resistor **R23** and a 2.2 nF capacitor **C19** arranged in parallel between the output and the negative terminal of the op-amp **U4A**. The positive terminal of the op-amp **U4A** is also connected to ground. The op-amp **U4A** is powered by a 3.8V supply, which is provided by the controller **1001** and which is connected to a power input of the op-amp via a 100 Ω resistor **R17**. The power input of the op-amp **U4A** is connected to ground via a 10 nF capacitor **C14**.

[0204] The first thermocouple op-amp **U4A** is configured to receive a signal from the first thermocouple constantan wire **1804** at the point **P8** and provide amplify this signal for outputting to the controller **1001**. In this example, the op-amp **U4A** provides a gain of 107.63 to the input signal from thermocouple reference end **1804a**. This gain may be provided by selecting appropriate values for the resistors **R26**, **R27** and **R23** for example. In selecting a gain for the op-amp **U4A**, the range of temperatures to be measured by the thermocouple **183a**, the reference voltage produced by a J-type at that range of temperatures and a suitable voltage range to be provided to the controller **1001** may be taken into account. In one example, a thermocouple reference voltage of 23.228 mV corresponds to a temperature reading of 425° C. To amplify this reference voltage to a value of 2.5V to provide to the controller **1001** requires a gain of 107.63, as provided by the example component values shown in FIG. **19**. An amplified first thermocouple voltage **TC1** is provided to the controller **1001** at a point **1901** which is connected to the output of the op-amp **U4A** via a 1 k Ω resistor **R22**. The point **1901** is connected to ground via a 100 nF capacitor **C21**. From the voltage signal **TC1** provided at the point **1901** the controller **1001** is configured to determine a temperature value measured by the first thermocouple **183a**, corresponding to the temperature of first susceptor zone **132a**.

[0205] The second thermocouple **183b** is, as described above, also a J-type thermocouple and provides a second thermocouple reference voltage **TC2** to the controller **1001** at a point **1902** in the

same manner as the first thermocouple reference voltage TC1 is provided to the controller **1001**. This allows the controller **1001** to determine a temperature for the second susceptor zone **132b** in the same manner as described for the first thermocouple **183b**. It will be understood that a plurality of resistors R37, R38, R39, R40 and capacitors C30, C31, C32 connected to the second op-amp U4B provide the equivalent function as the resistors R22, R23, R26, R27 and capacitors C19, C20, C21 described above in connection with the first thermocouple op-amp U4A.

[0206] In addition to providing to the controller **1001** the amplified thermocouple voltages TC1, TC2, the apparatus **1900** is also configured to provide a safety feature in the event that a temperature measured by either of the thermocouples **183a**, **183b** exceeds a given threshold. That is, the output of the first thermocouple op-amp U4A is connected to a negative terminal input of a first precision comparator U3A (component type AS393). A positive input of the precision comparator U3A is connected to a 2.5V signal via a 3.3 kΩ resistor R42 and to ground via a 10 kΩ resistor R43. The precision comparator U3A is configured such that if the amplified thermocouple voltage received from the op-amp U4A indicates a temperature exceeding a particular value, the first precision comparator U3A outputs a stop signal /FF RST to stop heating of the susceptor **132** by the induction heating circuit **600**. In examples, the signal /FF RST is sent to the flip-flop **622** input RST, to reset the output value Q of the flip-flop **622**, in the manner described above with reference to earlier figures, to thereby stop heating by the circuit **600**. In one example the threshold temperature value at which the apparatus **1900** is configured to send the signal to stop heating by the circuit **600** is 280° C.

[0207] A second precision comparator U3B performs an equivalent function for the second thermocouple **183b**, receiving an input at its negative terminal from the second thermocouple op-amp U4B and being configured to send the signal /FF RST if the received signal from the op-amp U4B indicates that the second thermocouple **183b** is measuring a temperature exceeding the threshold value.

[0208] A third comparator U6 is also included in FIG. **19** and provides for another mechanism by which the signal /FF RST may be sent to stop heating of the susceptor **132**. The comparator U6 receives a signal COIL TEMP from a thermistor (not shown) located on the insulating member **128** which indicates a temperature at the outside of the insulating member **128**, proximate the coils **124**, **126**. The comparator U6 is configured to compare this signal COIL TEMP to a threshold value and provide a signal, in this example a signal /FF RST to the flip-flop **622** to stop operation of the circuit **600** to heat the susceptor **132** if the comparator U6 determines that the value of the temperature represented by COIL TEMP is too high.

[0209] It will be appreciated that for accurate measurement of the temperatures of the susceptor zones **132a**, **132b**, each measurement ends (**1706**, **1710** in the example of FIG. **17**, or **1806**, **1810**, **1805** in the example of FIG. **18**) should be at a temperature which is substantially the same as the respective temperature of the zone **132a**, **132b** for which the temperature is being measured. As such, good thermal conductivity between the susceptor **132** and the thermocouples **183a**, **183b** should be maintained.

[0210] To describe this with reference to an example, it will be appreciated that if, for example, the attachment point **1806** of the first constantan wire **1804** were to become detached from the first susceptor zone **132a** then thermal conduction between that susceptor zone **132a** and the first thermocouple **183a** would be lost or at least significantly reduced. In such a circumstance, the first thermocouple **183a** may, for example, remain detached from but within close proximity of the susceptor **132** (and may continue to be heated by the susceptor **132** to some extent by e.g. radiative or convective processes). However, since heat would no longer be properly conducted between the susceptor **132** and the detached first thermocouple **183a**, the temperature of the first susceptor zone **132a** (which the first thermocouple **183a** is intended to measure) and the detached thermocouple wire **1804** may diverge as the first susceptor zone **132a** is heated. In this circumstance, the first thermocouple **183a** would not, for example, allow the controller **1001** to accurately determine

increasing temperature of its corresponding first susceptor zone **132a** as said zone **132a** is heated. If heating of the first susceptor zone **132a** were to continue in this manner without its temperature being properly measured by use of the first thermocouple **183a**, a danger of overheating of the susceptor **132** may arise. It will be appreciated that the same principle described above also applies to the second susceptor zone **132b** in the circumstance that the second thermocouple **183b** were to become decoupled therefrom.

[0211] In one example, the device **100** is configured to cut off heating of the susceptor **132** in the event of an indication that either of the susceptor thermocouples **183a**, **183b** is not correctly measuring the temperature of the susceptor **132**. This may be advantageous in that a safety feature is provided which allows cutting off of power to heat the susceptor **132** when the controller **1001** determines that the temperature of the susceptor **132** is not being properly measured.

[0212] In one example, as described above in relation to the induction heating circuit **600** of the device **100**, a power supplied to the induction heating circuit for heating of the susceptor **132** may be determined by measuring the DC voltage supplied by the DC supply and the DC current supplied by the DC supply and calculating a product of these values to obtain a power. From this measurement of power, the controller **1001** may be configured to determine an amount of energy supplied to the circuit over a given period of time. As above, the controller **1001** is also configured to receive temperature measurements from the thermocouples **183a**, **183b**. The controller **1001** is configured to determine a change in temperature measured by the thermocouples **183a**, **183b** over the given period of time. In one example then, the controller **1001** may determine the amount of energy supplied per 1°C . rise in temperature measured by one of the susceptor thermocouples **183a**, **183b**. For example, the controller **1001** may determine a ratio of the amount of energy supplied to the first susceptor zone **132a** over a given time period to the increase in the temperature measured by the first thermocouple **183a** over that time period. Similarly, the controller **1001** may determine the amount of energy supplied to the second susceptor zone **132b** per 1°C . rise in temperature measured by the second thermocouple **183b**—in examples this amount of energy per 1°C . rise is measured temperature may differ to that for the first zone due to differing properties of the first and second inductors **124**, **126**.

[0213] In one example, during setup of the device **100** for example, the amount of energy supplied to a given one of the inductors **124**, **126** for heating the corresponding susceptor zone **132a**, **132b** of that inductor which results in a measured temperature rise by the corresponding thermocouple **183a**, **183b** of a given amount may be determined. That is, through testing or otherwise, it may be determined that in normal operation when, for example, the first thermocouple **183a** is properly attached to the first susceptor zone **132a**, on average, supplying 400-1000 mJ, or around 500 mJ to the first inductor **124** results in a temperature rise measured by the first thermocouple **183a** of 1°C . That is, the ratio of energy supplied to measured increase in temperature may be 400-1000 mJ/ $^{\circ}\text{C}$., or around 500 mJ/ $^{\circ}\text{C}$. In another example, it may be determined that at any point during normal operation of the device, the maximum value of the ratio of energy supplied to the first inductor **124** to rise in temperature measured by the first thermocouple **183a** is 500 mJ/ $^{\circ}\text{C}$.

[0214] However, if, as described above, the first thermocouple **183a** becomes thermally decoupled from the first susceptor zone **132a**, then the amount of energy supplied to the first inductor **124** which results in a 1°C . temperature rise being measured by the first thermocouple **183a** may significantly increase. For example, through testing it may be found that if the first thermocouple **183a** is decoupled from the first susceptor zone **132a** then the ratio of the energy, as measured in mJ, supplied to the first inductor **124** to the increase in temperature measured by the first thermocouple **183a** may be around a very high number, such as 40,000 to 100,000 J/ $^{\circ}\text{C}$. That is, if the first thermocouple **183a** becomes detached from the susceptor zone **132a** and is no longer properly measuring its temperature, then the thermocouple may record only small increases in temperature, despite the susceptor **132a** being heated by the inductor **124**. The controller **1001** may thus determine that if the ratio of energy supplied to heat the first susceptor zone **132a** to the

temperature rise measured by the first thermocouple **183a** is greater than a given amount, then the first thermocouple **183a** may no longer be correctly measuring the temperature of the first susceptor zone **132a**, e.g. the first thermocouple **183a** may have become detached from the susceptor **132**. Similarly, the controller **1001** may determine that if a ratio of energy supplied to heat the second susceptor zone **132b** to the temperature rise measured by the second thermocouple **183b** is greater than a given amount, then the second thermocouple **183b** may no longer be correctly measuring the temperature of the second susceptor zone **132b**.

[0215] In this example, the controller **1001** is configured to stop the supply of power to heat the susceptor **132** if the ratio of energy supplied to heat the susceptor **132** to the temperature rise measured by one of the first and second thermocouples **183a**, **183b** is greater than a predetermined amount. Cutting-off of the power for heating the susceptor **132** in this manner may act as a safety feature which helps prevent over-heating of the susceptor **132** where one (or both) of the thermocouples **183a**, **183b** has become detached from the susceptor **132** or is otherwise no longer providing accurate susceptor temperature measurements. The controller **1001** may, for example, stop the supply of energy to heat the susceptor **132** by, deactivating the circuit **600** by use of a signal sent to the control arrangement of the induction heating circuit **600**. In another example, the controller **1001** may take another control action in response to determining that the ratio is greater than the predetermined amount. For example, the controller **1001** may reduce the power supplied to the circuit **600** to heat the susceptor **132**. For example, the controller **1001** may reduce the control voltage **1031** to reduce power supplied to heat the susceptor **132**.

[0216] In one example, an expected measured value of the ratio of energy to heat one of the susceptor zones **132a**, **132b** to the temperature rise measured by the corresponding thermocouple **183a**, **183b** in normal operation may be around 500 mJ/° C. and the controller **1001** may be configured to stop the supply of power to heat the susceptor **132** if it determines that the ratio has become greater than or equal to a example value from 2000 mJ/° C. to 4000 mJ/° C. As such, in this example, a margin, in this case of around 1500 mJ/° C. to 3500 mJ/° C. above the expected value is provided which may allow for minor increases of the ratio above the expected level to occur without the controller **1001** taking a control action such as cutting off the power supply.

[0217] In examples, during operation of the device **100** a value for the ratio between the energy supplied to heat the susceptor **132** to the measured increase in temperature sensed by one of the thermocouples **183a**, **183b** may be determined at predetermined intervals. The predetermined intervals may be, for example, the same as those used for controlling the induction heating circuit **600**, such as those described above, and may in one example be of length 1/64s. Where one of the thermocouples **183a**, **183b** has become detached from the susceptor **132**, it has been found that in one example a ratio of 4000 mJ/° C. for that thermocouple is exceeded within around 0.5 s. As such, in this example where the cut-off ratio is set at 4000 mJ/° C., the supply of power may be cut-off within around 0.5 s, and thus substantial overheating of the susceptor **132** may be avoided.

[0218] FIG. 20 shows a flowchart representation of an example method **1400** of controlling the device **100**. In an example the method **1400** is performed by the controller **1001**. At block **1402** power is supplied from the power supply to heat the susceptor **132**. At block **1404** an energy ΔE supplied to the heating circuit to heat the susceptor **132** over a given time period is determined. As described above, at any one time, power is only supplied to one or the other of the first inductor **124** and the second inductor **126** during the given time period. As such, the power measured at block **1404** is a power supplied to the one of the first inductor **124** and second inductor **126** which is active during the given time period. This may be determined as described above, for example, by measuring a DC voltage and a DC current supplied to the heating circuit by the DC supply and calculating a product of these values. For example, an energy used over the time period may be determined by determining an average power over the time period and multiplying this by the duration of the time period, e.g. of 1/64s.

[0219] At block **1406** a change in temperature ΔT measured by one of the susceptor thermocouples

183a, **183b** over the time period is determined. The temperature change ΔT will in examples correspond to the temperature change measured by the thermocouple corresponding to the inductor to which power was supplied during the given period. That is, for example, if during the given time period power is supplied to the first inductor **124**, then the energy ΔE is the energy supplied to the first inductor **124** during the given time period and the temperature change ΔT is the temperature change recorded by the first thermocouple **183a** during the given time period. At block **1408** a ratio $\Delta E/\Delta T$ of the energy ΔE supplied over the time period to the change in the temperature ΔT measured by the respective susceptor thermocouple **133a**, **133b** over the given time period is determined. At block **1410** the ratio $\Delta E/\Delta T$ is compared to a cut-off value. The cut-off value is a threshold value for $\Delta E/\Delta T$ which may be predetermined during setup of the device **100** as discussed above. The cut-off value may be greater than an expected value for the ratio $\Delta E/\Delta T$ when the device **100** is operating normally. The cut-off value may be greater than the expected value by a given margin, e.g. a margin of around 500 mJ/° C., or around 1000 mJ/° C.

[0220] If, at block **1410** the controller **1001** determines that the determined value for the ratio $\Delta E/\Delta T$ is less than the cut-off value then the method returns to block **1402** and the controller **1001** continues to allow power to be supplied to heat the susceptor **132** for another instance of the given time period. If, however, at block **1410** the controller **1001** determines that the ratio $\Delta E/\Delta T$ for the given time period is greater than or equal to the cut-off value then the method proceeds to block **1412**.

[0221] At block **1412** the controller **1001** determines if the inductor **124**, **126** has been active to heat the susceptor **132** for a predetermined length of time, of for example 0.2-1 s or around 0.5 s. The predetermined length of time may, for example comprise a number of consecutive predetermined number of time periods. For example, at block **1412** the controller **1001** may determine if the circuit **600** has been active to heat the susceptor **132** for 32 consecutive time periods of 1/64s, totaling a length of time of 0.5 s. If the answer is “no” then despite the determined ratio $\Delta E/\Delta T$ at block **1410** being above the cut-off value the method returns to block **1402** and the controller **1001** continues to allow heating of the susceptor **132**. The step performed at block **1412** allows the method **1400** to take into account that when the controller **1001** is operating to maintain the temperature of e.g. the first susceptor zone **132a**, rather than acting to increase the temperature of said zone **132a**, energy may be supplied to heat the zone **132a** while relatively small increases in temperature are recorded by the first thermocouple **183a**. Therefore, a value for $\Delta E/\Delta T$ above the cut-off value may be obtained which does not necessarily indicate a fault with the operation of the thermocouple **183a** to measure the temperature of the first susceptor zone **132a**.

[0222] In another example, the predetermined period over which ΔT and ΔE are calculated may be the same as the length of time used at block **1412**. That is, in one example, ΔT is the change of measured temperature over a 0.5 s period and ΔE the energy supplied over that 0.5 s period. In this example, therefore, at block **1412** the controller **1001** determines if the circuit **600** has been active to heat the susceptor **132** for the entirety of the previous 0.5 s time interval and if the answer is “yes” the controller **1001** proceeds to block **1414**. In this example, it will be understood that during the 0.5 s period the controller **1001** may determine once every 1/64s which of the inductors **124**, **126** if either should be active to heat the susceptor **132** and thus block **1412** may amount to the controller **1001** determining that one of the inductors **124**, **126** was active for each of the 32 previous 1/64s intervals.

[0223] At block **1414**, the controller **1001** cuts off the supply of power to heat the susceptor **132**. This acts as a safety mechanism which may prevent over-heating of the susceptor **132** in the event that one or both of the thermocouples **183a**, **183b** is not properly measuring changes in temperature of the susceptor **132** for a given length of time.

[0224] As described in examples above, the controller **1001** is configured to implement the described safety feature by determining if a predetermined criteria is satisfied based on a temperature of the susceptor **132** and the energy supplied to the circuit **600**. However, it will be

appreciated that in other examples, the controller **1001** may determine whether to take a control action such as stopping the supply of energy to the heating circuit based on one or more predetermined criteria using a different temperature measured in the device **100**. For example, the above-described method of determining, based on a comparison between a measured increase in the temperature and an amount of energy supplied to the heating circuit, if a criteria is met which indicates a fault in measuring a temperature in the device **100** may be used elsewhere in the device **100**.

[0225] As described above, the controller **1001** of the example induction heating device **100** is configured to perform a method of stopping supply of energy to heat a susceptor if one or more predetermined criteria that are indicative of a fault with measurement of a temperature of the susceptor are satisfied. However, in other examples, the above-described method may be applied by a controller in a different type of aerosol generating device than a device which comprises an inductive heating circuit. For example, the above-described example method may be applied in an aerosol generating device comprising a heating circuit which is not an inductive heating circuit. In such an example the heating circuit may comprise one or more resistive heating elements for generating heat to heat an aerosol generating material when a current is passed through said resistive heating elements. In an example a temperature sensing arrangement may comprise temperature sensors for sensing temperatures of one or more such a resistive heating elements and such temperature sensors may be attached to or in close proximity to said heating elements during normal use. The controller of such a device may be configured to determine a fault with the arrangement, such as detaching of one of the temperature sensors from the heating element, by methods described above with reference to FIG. **20**.

[0226] Examples above have described the controller **1001** determining a predetermined criteria which is dependent on a measured increase in temperature of the thermocouples **183a**, **183b**. In other examples, the controller **1001** may, for example, determine that a fault has occurred with the temperature sensing of the susceptor **132** by the thermocouples **183a**, **183b** if the controller determines that energy has been supplied to heat the susceptor **132** over a given period while a reduction in temperature has been measured by the thermocouples **183a**, **183b** over that same period.

[0227] Certain methods described herein may be implemented by way of non-transitory computer program code that is storable on a non-transitory storage medium. For example, in certain examples, the controller **1001** may comprise a non-transitory computer readable storage medium comprising a set of computer-readable instructions stored thereon and a processor to perform a method described herein when executed by the controller **1001**. The controller **1001** may comprise one or more processors. For example, in some examples, as described above, the controller **1001** is a programmable micro-processing unit. The controller **1001** may comprise a storage medium comprising a set of machine readable instructions, e.g. in the form of computer code, which when executed by the controller **1001** cause a method described herein to be performed.

[0228] While some of the example circuits described herein make use of silicon FETs for certain switching functions other suitable components may be used in place of such FETs. For example, components comprising wide bandgap materials such as silicon carbide, SiC, or gallium nitride, GaN, may be used. Such components may in some examples be FETs but in other examples may be high electron mobility transistors (HEMT). Such components may be faster and have higher breakdown voltages than silicon FETs which may be advantageous in some examples.

[0229] The above embodiments are to be understood as illustrative examples of the invention. Further embodiments of the invention are envisaged. It is to be understood that any feature described in relation to any one embodiment may be used alone, or in combination with other features described, and may also be used in combination with one or more features of any other of the embodiments, or any combination of any other of the embodiments. Furthermore, equivalents

and modifications not described above may also be employed without departing from the scope of the invention, which is defined in the accompanying claims.

Claims

1. Apparatus for an aerosol generating device, the apparatus comprising: a heating circuit for heating a heating arrangement, the heating arrangement arranged to heat an aerosol generating material to thereby generate an aerosol; a temperature sensing arrangement for sensing a temperature of the device; and a controller for controlling a supply of energy to the heating circuit, wherein the controller is configured to: determine a characteristic that is indicative that energy is being supplied to the heating circuit over a given time period, and determine a change in temperature sensed by the temperature sensing arrangement over the given time period; and take a control action if, based on the determined characteristic and the increase in temperature sensed by the temperature sensing arrangement over the given period, the controller determines that one or more predetermined criteria that are indicative of a fault with the temperature sensing arrangement are satisfied.
